



Research paper

A domain-adapted NLP pipeline for human factors classification of maritime accidents

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ABSTRACT

Human error continues to account for most maritime accidents, highlighting persistent challenges in safety management and organisational learning. Although post-incident investigations generate rich narrative data, extracting meaningful insights is hindered by manual coding, subjective interpretation, and inconsistent application of human-factors taxonomies. Despite increasing interest in applying NLP to safety analysis, no existing approach provides a taxonomy-ready, domain-adapted framework capable of reliably classifying fine-grained human and organisational factors in maritime accident narratives. This study presents a natural language processing (NLP) and machine learning (ML) framework to automate the identification of human and organisational contributory factors in maritime accident reporting using an adapted SHIELD taxonomy. The approach analyses individual causal and contributory text segments extracted from accident investigation narratives, which are classified into human and organisational factor codes. Central to the approach is SeaSafeBERT, a domain-adapted BERT model fine-tuned on an international corpus of maritime investigations and combined with TF-IDF lexical features to support structured, multi-level classification. To address sparse taxonomy classes, we developed SHIELD-NLP, a consolidated coding scheme preserving SHIELD's intent but ensuring sufficient class representation across a dataset of 330 maritime accident investigations and enabling classification into 53 SHIELD-NLP codes. The system achieved strong performances for many operationally important SHIELD-NLP codes, with several reaching F1 scores above 0.80. Association rule mining further revealed consistent interaction patterns between perceptual, cognitive, and organisational factors across incident types. Together, these contributions represent, to our knowledge, one of the first domain-specific maritime safety language models and integrated pipelines for automated SHIELD-based analysis, demonstrating the potential of AI-enabled methods to accelerate incident learning, improve code consistency, and strengthen proactive safety management in maritime operations.

1. Introduction

The maritime industry is crucial to global trade, responsible for over 80% of all world exports and imports, including nearly all food and manufactured goods (UNCTAD, 2025). The scale and interconnectedness of global shipping mean that even a single major incident can have wide-ranging impacts on international supply chains, coastal communities, and marine ecosystems. As shipping volume continues to grow and vessel operations become more complex, safety management must evolve beyond compliance-based systems to data-driven, predictive approaches.

Despite significant regulatory changes aimed at improving safety, maritime incidents remain a persistent challenge (Rothblum, 2000).

Although using both reactive and proactive approaches to help increase safety and drive change to culture and procedures, maritime incidents remain a persistent concern (EMSA, 2022). The effects of these incidents can often be detrimental to the environment and the human element, where these incidents often cause injuries and in worst cases, fatalities.

It is estimated that up to 80% of these incidents could be mitigated through improved human intervention and the application of human factors principles in both design and operations (Wagenaar and Groeneweg, 1987). Recognising the human element as a system component rather than an isolated case has become central to contemporary safety thinking. Frameworks such as Reason (1990) "Swiss cheese" model and SHIELD taxonomy emphasise that human error is rarely a simple, isolated occurrence; it is often the result of complex interactions involving

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multiple contributing factors. These factors, referred to as “human factors,” encompass environmental, organisational, and individual characteristics, such as communication breakdowns, fatigue, organisational culture, human-machine interaction, and physical health. Understanding which factors contribute to human error is vital for developing regulations and procedures that increase safety and prevent future incidents. As such, the effective and efficient processing of data containing insights into human errors and the root causes of maritime incidents is of paramount importance. These incidents, though regrettable, present a unique opportunity for the industry to learn invaluable lessons by refining existing procedures and enhancing equipment designs to rectify the faults exposed by these unfortunate occurrences (Ashraf, 2014).

In the aftermath of an incident, investigations are conducted to determine the causes and provide recommendations for future prevention. These investigations generate free-text reports detailing what happened, why it happened, and what can be done to avoid recurrence. Maritime administrators and insurers collect thousands of these investigation reports per year. The process of reading, coding and analysing these documents has become unsustainable, as traditionally the extraction of information from these reports, through human factors taxonomies, has been a labour-intensive and costly endeavour, primarily conducted by human factors experts. Analysts also face challenges, including inconsistent language use, varying report structures, and subjective interpretations of causal factors. These limitations slow the feedback loop between incident occurrence and prevention action, reducing the system's capacity to learn in real time.

However, recent advances in Natural Language Processing (NLP) have led to changes, where there is now new potential to automatically extract factors and root causes from these reports (Louis et al., 2024; Ma and Chen, 2024). Although these approaches are typically domain-agnostic and not aligned with structured human-factors taxonomies. In other transport domains, such as Aviation and Rail (Lázaro et al., 2025; Williams and Betak, 2018) NLP has been proven to successfully categorise causes of incidents into predefined taxonomy structures. These tools can process large volumes of data far beyond human capacity, offering faster, more consistent analysis. However, the maritime sector has lagged in utilising these new technological developments for this purpose. This slower adoption stems from data fragmentation, variations in reporting quality, and the absence of a universally adopted human factors taxonomy for maritime use. As a result, insights from incident investigations are often underutilised, remaining locked within textual reports.

To structure this process, the SHIELD taxonomy provides a comprehensive framework for classifying human, organisational, and environmental factors in maritime incidents. The framework was developed to mirror taxonomies such as HFACS and SHEL but adapted for maritime and aviation contexts, allowing for systematic comparison across cases (Navas de Maya et al., 2021). However, SHIELD was designed for manual application, and its fine-grained categories are not immediately suitable for automated text analysis. Adapting the taxonomy for machine-learning use, therefore, represents a critical step towards scalable, consistent human-factors classification.

As a result, there remains a clear gap in the literature for maritime-specific, domain adapted NLP framework that supports automated, fine-grained classification of human and organisational factors using an established human-factors taxonomy. Addressing this gap is essential for enabling scalable, consistent learning from maritime accident investigations.

This study addresses this gap by using Artificial Intelligence (AI) techniques, specifically NLP and ML methods, to automate the classification of human and causal factors found in incidents. The approach includes (i) restructuring SHIELD into a machine-learning-ready taxonomy (SHIELD-NLP), (ii) developing a domain-specific language model (SeaSafeBERT), adapted from BERT to capture maritime investigative language, (iii) implementing a classification pipeline that issues a hybrid class imbalance methodology to predict SHIELD-NLP codes from

narrative text, and (iv) applying association rule mining to analyse the interactions between causal factors. The resulting system generates structured causal factors at scale and supports downstream analysis of patterns and co-occurring risk.

The primary objective of the paper is to evaluate an AI-driven framework for the automated identification of human and causal factors in maritime incidents using the SHIELD taxonomy. To achieve this objective, the study pursues the following methodological and analytical aims. The study's methodological objectives are to:

- Restructure the SHIELD taxonomy into a machine-learning-ready coding scheme (SHIELD-NLP) that preserves its conceptual intent while addressing class imbalance,
- Develop a domain-adapted language model (SeaSafeBERT) tailored to maritime accident investigation narratives,
- Design and evaluate a hierarchical, multi-level classification pipeline for predicting SHIELD-NLP codes from free-text reports.

Beyond its methodological objectives, the research aims to:

- Examine interaction and co-occurrence patterns between human and organisational factors using association rule mining,
- Assess the potential of the proposed framework to support scalable organisational learning by increasing the speed, consistency, and analytical depth of human-factors assessment.

This study makes the following novel contributions to maritime safety research by introducing a domain-adapted NLP framework for maritime accident analysis. Its key contributions include the development of SeaSafeBERT, a maritime-specific language model, the construction of SHIELD-NLP, a machine-learning-ready adaptation of SHIELD taxonomy, an integrated hierarchical classification pipeline for automated human-factors coding, and the application of association rule mining to uncover interaction patterns between contributory factors, supporting scalable organisational learning in maritime safety management.

The remainder of this paper is structured as follows. Section 2 provides a comprehensive literature review covering maritime human-factors taxonomies, NLP-based incident analysis in safety-critical domains, and challenges associated with taxonomy-driven classification. Section 3 outlines the methodology used to construct SHIELD-NLP, develop SeaSafeBERT, and design the multi-level classification pipeline. Section 4 presents the classification and association rule mining results, and Section 5 discusses these results, their implications for maritime safety practice, and recommendations for future work. Finally, Section 6 concludes by highlighting the contributions of the paper.

2. Literature review

Although numerous human factors taxonomies have been developed to structure the analysis of safety incidents, their manual application remains highly time-consuming and inconsistently used in practice. In recent years, the utilisation of NLP techniques for incident analysis has gained momentum to address this challenge. Although usage within other industries has been growing, the application of NLP in the maritime domain is still emerging, with limited studies exploring its potential. This review will explore the methods and approaches used in maritime and other safety-critical domains to help reduce the labour-intensive burden of the analysis and to strengthen the methodological foundations for extracting structured insights from narrative safety data.

2.1. Maritime human factors taxonomies

Human-factors taxonomies provide a framework for organising contributory factors, enabling investigators to compare cases, spot patterns, and target interventions. Across maritime-focused studies, there

are many taxonomies used, like HFACS, FRAM, AcciMap, TRACER and system-theoretic approaches such as STAMP/CAST. There are also newer hybrid models, such as SHIELD and SHARE, which have also been effectively used.

HFACS remains the most commonly used method rooted in Reason's model. The model is known for being clear, low in training burden, and compatible with qualitative add-ons. However, limitations are equally well-reported, such as its inherent subjectivity, inter-coder variability, and that thin reporting of organisational and cultural contributors in many investigation records constrains the coding ability (Galieriková, 2019; Mazaheri et al., 2015).

FRAM and AcciMap extend analysis to system interactions but are time-consuming and heavily analyst-dependent (Filho et al., 2019; Salihoglu and Bal Beşikçi, 2021). TRACER focuses on cognitive processes in human-machine interfaces (Graziano et al., 2016), and STAMP-CAST models complex socio-technical control failures (Ceylan et al., 2021), yet both require specialised expertise and are rarely applied at scale.

Across these frameworks, several recurring limitations have been reported in the literature. Analyses are often described as subjective due to their reliance on expert judgement, which in turn means it can be difficult to resource across large volumes of reports (Paul et al., 2012; Underwood and Waterson, 2013). Consequently, multiple authors have argued for the development of more structured, domain-generic taxonomies that can be integrated with data-driven approaches (Stroeve et al., 2023b).

The Safety Human Incident and Error Learning Database (SHIELD) taxonomy, developed during the SAFEMODE project (Ramirez Marengo et al., 2021), directly addresses these challenges. Built from NASA-HFACS and HERA, SHIELD offers a domain-independent but systematically defined framework (Navas de Maya et al., 2021). Evaluation of the method showed that SHIELD enabled identification of over twice as many causal factors as HFACS-based analyses, reflecting greater sensitivity to latent organisational and system issues (Stroeve et al., 2023b). Its concise definitions, minimal training requirement, and consistent coding make it particularly suitable for digital implementation and NLP-based automation.

Despite the breadth of available human-factors frameworks, none of the existing taxonomies strike a balance between the level of granularity required for meaningful causal analysis and statistical proprieties needed for machine-learning applications. Highly detailed frameworks, such as SHIELD, offer nuanced causal categories suitable for expert investigation, but this granularity produces extremely sparse and long-tailed label distributions when applied to real-world incident datasets. On the other hand, more aggregated taxonomies reduce sparsity but become too coarse to support fine-grained causal reasoning or automated safety intelligence. As a result, there is currently no taxonomy that is simultaneously rich enough for human-factors analysis and sufficiently well-distributed to be learnable by supervised NLP models.

2.2. Challenges in applying taxonomies in machine learning

While numerous studies have applied NLP to incident analysis, fewer have focused on taxonomy-based causal factor extraction, where text is mapped directly to structured taxonomic categories. Early work in aviation and healthcare demonstrated the feasibility of using a supervised model to classify incident narratives into HFACS or SHEL categories (Madeira et al., 2021; Perboli et al., 2021; Tanguy et al., 2016). These studies showed that taxonomy-guided labelling improved interpretability and alignment with expert analysis, but small, imbalanced datasets and extensive manual annotation requirements limited them.

In the process and construction industries, similar taxonomy-aligned approaches have been tested. For example, Sahoo et al. (2024) used rule-based extraction followed by logistic regression to assign causal categorisation, while Yang and Demichela (2023) combined rule logic with supervised models to extract “who, when, why” dimensions from accident narratives. Both highlight that hybrid NLP-ML methods can

automate parts of causal classification while retaining transparency for auditors and investigators.

Deep learning has further advanced taxonomy-based analysis. Studies have shown that fine-tuned transformer models, such as BERT and RoBERTa, outperform traditional text representations in extracting contributory factors from structured safety reports (Kim et al., 2024; Oliäee et al., 2023). However, these models still require domain-specific fine-tuning to capture technical terminology and causal phrasing unique to safety narratives (Nielsen et al., 2024).

Although these studies demonstrate the potential of NLP for taxonomy-driven analysis, applying machine learning to large, fine-grained human-factors taxonomies remains challenging. Most taxonomies, including HFACS, HERA, and SHIELD, contain dozens of highly specific categories, many of which occur rarely in incident datasets. This produces a long-tailed label distribution in which a small number of common factors receive substantial representation, while many others appear only a handful of times. This sparsity limits the ability of supervised models to learn stable decision boundaries and leads to poor calibration, high variance in minority classes, and overfitting to spurious linguistic patterns (Wu and Mei, 2026).

In addition, safety taxonomies often contain overlapping or hierarchically related constructs, which increases the difficulty of direct mapping free-text narratives to discrete categories. These structural and statistical constraints mean that taxonomy-based NLP pipelines often require taxonomic simplification as well as domain-specific embeddings and resampling before reliable automated classification can be achieved.

2.3. Dealing with class imbalance and embeddings methods

A persistent challenge in applying NLP and machine learning to taxonomy-based incidents is the highly uneven distribution of contributory factors within safety datasets (Liu et al., 2025; Stroeve et al., 2023b). Rare but operationally important categories, such as subtle cognitive errors or latent organisational conditions, often appear only a small number of times, while a few common factors dominate the dataset. Studies in aviation, healthcare, and industrial safety consistently show that such imbalance severity degrades model performance, particularly recall for minority classes. To address this issue, various strategies have been explored in the literature, including class weighting (Baker et al., 2020), down-sampling the majority classes (Madeira et al., 2021), and synthetic oversampling using SMOTE (Ebrahimi et al., 2023; Luo et al., 2023). SMOTE, in particular, has become widely adopted for safety data because it generates plausible synthetic minority samples without disregarding real data. These methods have been shown to improve stability and minority-class detection, yet imbalance remains a fundamental constraint when taxonomies are fine-grained, and incident volumes are modest.

A second, closely related challenge concerns the specialised language used in safety reporting. Incident narratives often contain technical terminology, shorthand operational phrases, and domain-specific causal cues that generic language models do not capture well. Although transformer-based models such as BERT and RoBERTa outperform traditional embeddings in many safety-related NLP tasks, studies consistently highlight that domain adaptation, fine-tuning on incident corpora or embedding sector-specific terminology, is necessary to achieve reliable performance (Nielsen et al., 2024; Oliäee et al., 2023). To date, however, no publicly documented BERT-based language model has been specifically adapted for maritime accident investigation narratives, leaving a gap between generic transformer representations and the linguistic characteristics of safety reporting. Without such adaptation, contextual models may misinterpret critical operational terms or overlook low-frequency phrases that strongly map to taxonomy codes.

Additionally, across safety-oriented NLP studies, TF-IDF remains one of the most effective methods for representing discriminative lexical features. Research comparing embedding methods showed TF-IDF paired with linear classifiers frequently outperforms more complex

models when task performance hinges on specific terminology (Macedo et al., 2023). This robustness is particularly valuable in safety taxonomies, where subtle wording differences can distinguish between closely related causal categories.

Despite the demonstrated strengths of both contextual and lexical embedding methods, the reviewed literature shows that they have been applied independently rather than in combination. The potential benefits of combining contextual embeddings with sparse lexical representations remain underexplored in safety-specific, taxonomy-driven analysis.

2.4. Association rule mining in safety science

A small body of research has also explored statistical pattern-mining techniques to uncover causal structures in safety data. Association Rule mining (ARM) has emerged as a promising technique for identifying recurrent combinations of human, environmental, and organisational factors. For example, Jin et al. (2025) applied ARM to Arctic maritime incidents, introducing a domain-specific lexicon and proposing a modified interestingness measure that improved the detection of meaningful causal relationships. Other studies have combined ARM with complementary analytical frameworks, such as Sahoo et al. (2024), who integrated LDA topic modelling, ARM, and fault tree analysis to uncover accident chains in steel plant incidents.

Within maritime safety, hybrid approaches blending ARM with systemic models have also been explored for instance Bairami-Khankandi et al. (2025) used CAST alongside ARM and BN-based trend analysis to reveal cross-vessel accident patterns. Together, these studies show that ARM is well-suited to analysing taxonomic or multi-factor datasets, as it can reveal co-occurrence patterns and latent causal pathways without requiring complex semantic extraction. However, its application to maritime human-factors taxonomies remains limited, and no prior work has examined ARM in conjunction with NLP-driven classification pipelines such as those developed in the present study.

Overall, the reviewed literature highlights substantial progress in (i) NLP-based analysis of safety narratives, (ii) taxonomy-guided causal classification, and (iii) pattern-mining approaches such as association rule mining. However, these strands remain largely separate in maritime research. Existing maritime NLP applications do not yet combine domain-adapted transformer models, class-imbalance-aware taxonomy restructuring, and structured causal pattern analysis within a unified pipeline. This fragmentation motivates the integrated framework proposed in this study.

3. Methodology

3.1. Data and taxonomy framework

3.1.1. SHIELD database

The foundation of this study is the Safety Human Incident and Error Learning Database (SHIELD). This structured framework was developed during the European Union-funded SAFEMODE project to systematically classify human errors and causal factors in maritime and aviation incidents. The Taxonomy integrates established concepts from safety science, including James Reason's Swiss Cheese Model, the Human Factors Analysis and Classification System (HFACS), and Human Error in Air Traffic Management (HERA) (Stroeve et al., 2023a).

The framework models accident causation as a multi-layered process, often referencing the human factors iceberg to suggest that causal factors extend far deeper than the surface-level observation. The taxonomy is organised hierarchically into four primary layers: Acts, Preconditions, Operational Leadership, and Organisational factors, which provide increasing depth of causal reasoning. Expert engagement was used to refine code definitions and classifications, ensuring the taxonomy clarity and practical applicability in investigative contexts.

3.1.2. Taxonomy adaptation (SHIELD-NLP)

The original SHIELD taxonomy contains 99 human-factor codes with substantial variation in class frequency, many of which appear only a small number of times in maritime incident reports. To prepare the taxonomy for supervised multi-label classification, we conducted a frequency analysis of all codes and applied a minimum support threshold of 20 positive instances for inclusion in the machine-learning schema. This threshold was selected to ensure that each label had sufficient representation from stable model training.

Codes with fewer than 20 examples were systematically reviewed. When two or more low-frequency codes described the same underlying human-factor constricted but differed only in operational context, they were merged into a single category. Codes that were both infrequent and conceptually distinct were removed from the machine-learning taxonomy, as they could not be reliably modelled. All proposed mergers and exclusions were reviewed to confirm that the revised taxonomy preserved the conceptual integrity and hierarchical structure of SHIELD.

The process resulted in a consolidated taxonomy, SHIELD-NLP, consisting of 53 machine-learning-ready labels. The multi-label structure of the original coding system was retained, allowing incidents to remain associated with multiple contributing factors.

More specifically, each sentence-level instance is represented using a binary multi-label encoding, where each dimension corresponds to one SHIELD-NLP code and multiple codes may be active simultaneously.

A detailed breakdown of the number of sentence-level instances per SHIELD-NLP label across the training, validation, and test splits is provided in Table A3 of the Appendix.

3.2. Model pipeline

This study developed a human-in-the-loop text classification pipeline designed to automatically map free-text causal statements from maritime incident reports to the predefined SHIELD-NLP codes. The unit of analysis in this study is the individual causal sentence extracted from an accident investigation report. Each report may yield multiple causal sentences treated as an independent instance while retaining a reference to its source report. Sentence-level instances inherit the expert-assigned SHIELD-NLP labels associated with the originating report, allowing each instance to be associated with multiple contributing factors in a multi-label representation.

The pipeline comprises five principal stages, shown in Fig. 1: (1) document ingestion and causal sentence extraction, (2) hybrid domain-specific language representation, (3) supervision classification with imbalance handling and probability calibration, (4) decision logic incorporating confidence thresholds and domain rules, and (5) model validation. This design supports both automated pre-classification and expert-validated refinement.

3.3. Dataset preparation

3.3.1. Causal sentence extraction

To maximise causal signal and reduce noise in long-form incident reports, we implemented a rule-based extraction mechanism to isolate sentences containing explicit causal information. This preprocessing step involved:

1. Sentence splitting and cue scoring: Each report was first segmented into sentences, a cue-based scoring mechanism was applied. Each sentence received a causal score based on curated causal cues (e.g. "because", "due to", "resulted in") and was penalised by non-causal suppressors (e.g. "introduction", "appendix", "references").
2. Length normalisation: False positives were reduced by applying length normalisation to very short sentences (<6 tokens) and very long sentences (>60 tokens)
3. Sentence selection: Sentences were retained if they exceeded a minimum causal score of 0.8, with a fallback to the top-N sentences

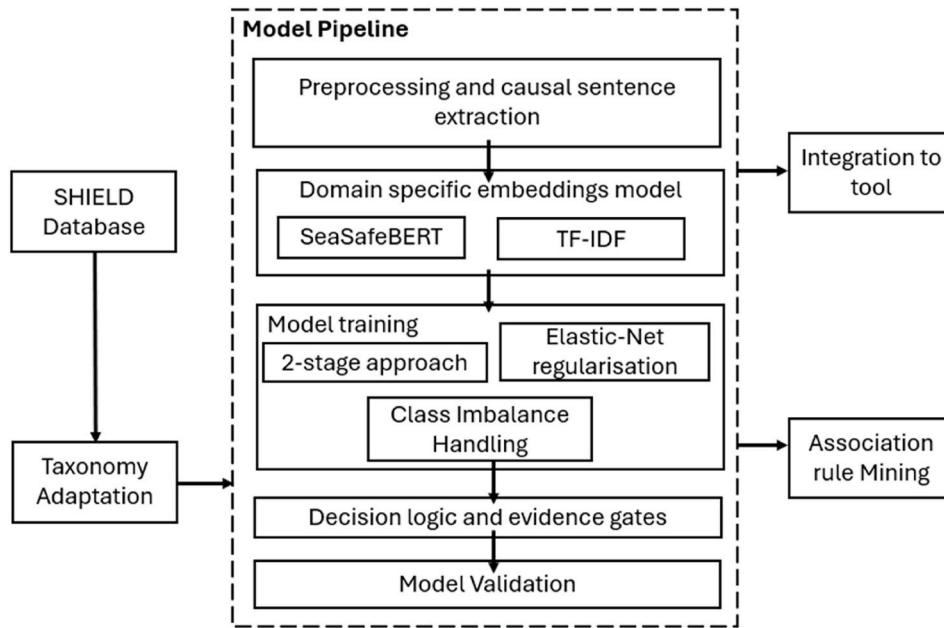


Fig. 1. Methodology workflow diagram.

(specifically, $N = 150$) by score to ensure sufficient material was extracted from each document.

3.3.2. Data splitting

After causal sentence extraction, the dataset was divided into training, validation and test sets using a 5-folds stratified splitting strategy to preserve label prevalence across splits. First, a global train-test split was performed, allocating 80% of the data to training and 20% to a held-out test set. The split was stratified by SHIELD-NLP code.

Validation data were then drawn exclusively from the training portion. For each one-vs-rest classifier corresponding to a SHIELD-NLP code, the training set was further split into a model-training subset (80%) and a validation subset (20%), again using stratification to maintain the positive-negative class balance. This resulted in an effective data allocation of approximately 64% training, 16% validation, and 20% testing.

To prevent information leakage, all resampling procedures (SMOTE), probability calibration, and decision-threshold tuning were performed using only the training and validation data. The held-out test set was not used during model fitting, resampling, calibration, or threshold selection and was reserved solely for final performance evaluation.

Splits were performed at the sentence level rather than the report level. While this may allow sentences from the same report to appear in different splits, the use of sentence-level units reflects the intended operational setting in which individual causal statements are classified independently.

3.4. Feature engineering

3.4.1. Domain adaptation (SeaSafeBERT)

To enhance performance on maritime classification tasks, a version of BERT was fine-tuned using the Masked Language Modelling (MLM) objective on a domain-specific corpus of accident reports. This adapted model is termed SeaSafeBERT.

The corpus was constructed from 250 maritime accident investigation reports, from the UK Marine Accident Bureau (MAIB) ($n = 100$),

Australia Transport Safety Bureau (ATSB) ($n = 50$), Transport Safety Board of Canada ($n = 50$) and the United States National Transportation Safety Board (NTSB) ($n = 50$). These authorities were selected because they provide detailed, narrative-based investigation reports with structured causal analysis and consistent documentation standards. Reports were converted to plain text, and only documents exceeding 500 tokens were retained, ensuring that actual reports rather than just summaries were included. The text was tokenised using a pre-trained bert-base-uncased tokenizer, which inputs padded or truncated to a fixed length of 128 tokens. The domain-adaptation corpus comprised approximately 3.6 million tokens.

The training was conducted using the HuggingFace Trainer API with the following key hyperparameters: batch size 8, 3 full epochs over the corpus, and a learning rate of $5e-5$. Fine-tuning BERT on this corpus allows the model to internalise the vocabulary, stylistic patterns, and semantic structures characteristics of maritime incident narratives, making SeaSafeBERT better suited than general-purpose language models for downstream tasks. Additionally, all experiments used a fixed random seed (42) to ensure reproducibility. Training was conducted on a CPU-based workstation and took approximately 2 h.

3.4.2. Hybrid embedding method

The proposed models consume sentence-level vector representations constructed via a hybrid embedding pipeline that combines semantic context from a transformer encoder (domain-specific BERT) with high-precision lexical cues from a sparse model (TF-IDF).

To retain sensitivity to domain-critical keywords (e.g. “handover”, “lookout”, “standing orders”), the SeaSafeBERT embeddings were concatenated with a sparse (Term Frequency-Inverse Document Frequency) TF-IDF lexical head. This hybrid feature vector balances semantic context with precise operational terminology. The TF-IDF model utilised term-frequency scaling, a minimum document frequency of ≥ 3 , a maximum document frequency cap of 0.95, and unigram and bigram token extraction with maximum 5000 features.

The final embedding for each sentence was therefore a hybrid vector combining the dense contextual semantics from SeaSafeBERT and the

high-precision lexical features from TF-IDF, the final dimensionality of the vector was 5768. This design helps preserve both semantic generalisation and sensitivity to operational vocabulary.

3.5. Hierarchical classification

The modelling process exploited the known hierarchical structure of the SHIELD taxonomy by implementing a two-stage hierarchical classification pipeline as an alternative to a flat model. The first stage used a single multi-class classifier, trained to predict the top-level Parent Category. In stage two, for each parent category, a separate OVR logistic regression model was trained on only the subset of the data positively labelled with that category. This model was responsible for predicting the specific SHIELD-NLP Code nested within its category. The final prediction for a specific SHIELD-NLP code was only output if both the stage one model predicted its Parent Category and the corresponding stage two model predicted the specific code. These stage two classifiers are applied conditionally based on stage one outputs. A SHIELD-NLP code is emitted only if both its parent category and corresponding stage 2 classifier exceed their respective decision thresholds, enforcing consistency with the SHIELD hierarchy while allowing multiple child codes to be predicted.

The classifier used was a One-vs-Rest (OVR) Logistic regression model with Elastic-Net regularisation (L1 ratio = 0.25). Logistic regression was selected because it provides transparent and interpretable decision boundaries, supports well-calibrated probabilistic outputs and remains robust. Elastic-Net regularisation was then applied to help mitigate multicollinearity and overfitting, which are common in high-dimensional text embeddings.

3.6. Imbalance handling

Then, to address the inherent imbalance in human-factors data, a multi-step strategy was employed: first, SMOTE was applied; next, class weighting and inverse-frequency sample weights were used to reinforce minority classes. Given the class distribution, SMOTE was not applied indiscriminately to all classes. Instead, a sensitivity analysis was conducted to determine the minimum number of positive instances required for stable synthetic oversampling. Classes with extremely low support can produce unstable synthetic neighbours, potentially introducing noise rather than improving generalisation.

To evaluate this, we compared model performance under different minimum-support thresholds for SMOTE eligibility, while maintaining class weights for these classes. Performance was assessed using macro and micro-averaged precision, recall and F1-scores on a validation split. The results indicated that applying SMOTE to only classes with at least 20 training instances improved the resulting model, balancing variance reduction for moderately represented classes with stability preservation for rare categories.

To improve prediction quality, probabilities were also calibrated using CalibratedClassifierCV. The method put preference on isotonic regression when sample size permitted, but used Platt scaling otherwise. In addition, cross-validation folds were chosen adaptively based on available positives.

3.7. Decision logic and evidence gates

The system outputs two prediction states for expert workflow: Auto (high-confidence automation) and Suggest (borderline cases requiring review). This logic is driven by dual prediction thresholds:

1. Fixed thresholds: a simple baseline was implemented with a fixed high-confidence threshold of $P \geq 0.5$ (Auto) and a low-confidence threshold of $P < 0.5$ and $P \geq 0.2$ (Suggest).
2. F1-Tuned Thresholds: The primary approach involved tuning the high threshold on the validation set to maximise the F1-score. The corresponding low threshold was then set using the following adaptive rule, where ϵ is a small positive margin and LOW_{min} is a minimum allowable value for the low threshold, and thr_{high} is the F1-optimised high-confidence probability threshold:

$$thr_{low} = \min(0.9 \cdot thr_{high}, \max(LOW_{min}, thr_{high} - \epsilon))$$

To further reduce implausible hits, especially for rare codes, evidence gates (domain rules) were applied after probability scoring. These gates included required keywords, veto keywords and post-probability filters.

3.8. Model validation

Model performance was assessed on the held-out test set using evaluation metrics that are appropriate for imbalanced, multi-level safety data. Traditional correctness-based measures such as overall accuracy were not used as primary indicators of performance because they can give a misleading impression in domains where many categories are rare. In the SHIELD-NLP taxonomy codes appear infrequently meaning a model could achieve a superficially high accuracy simply by predicting that these codes never occur, while still failing to identify safety issues.

To obtain a more meaningful assessment, we report precision, recall, F1-score and ROC-AUC for each code. Precision reflects how often the model's positive predictions are actually correct. This is important for operational use, as excessive false alarms would reduce analysts' trust in automated suggestions. Recall measures how effectively the model identified all instances of a given human-factor category. High recall is particularly important in safety analysis, where missing a genuine contributing factor can have greater consequences than flagging extra ones. The F1 score then combines precision and recall into a single measure, summarising the model's ability to balance these two requirements. It is well-suited for tasks where both over-prediction and under-prediction carry operational implications. ROC-AUC evaluates the model's ability to discriminate between positive and negative examples across all possible thresholds, independent of any specific cutoff. This provides insight into the underlying separability of the classes, even when threshold metrics such as F1-score may be affected by class imbalance. Together, these metrics offer a comprehensive and reliable picture of model performance in a multi-label, imbalanced safety dataset.

All preprocessing, resampling (SMOTE), class weighting, probability calibration, and threshold optimisation were confined strictly to the training and validation subsets within each split. The held-out test set remained untouched until the final evaluation.

3.9. Association rule mining (ARM)

Association Rule Mining (ARM) was applied to identify recurrent patterns of co-occurring contributory factors within the SHIELD-NLP taxonomy. The objective was to uncover statistically meaningful relationships between human, environmental, organisational, and supervisory factors to examine whether these relationships differed across accident types. ARM was conducted using the Apriori algorithm, which identifies frequent items and derives association rules that meet user-defined support, which was set to 5% for this analysis. This means that the rule had to occur in at least 5% of cases to be shown in the

network. This was selected to balance the interpretability and statistical reliability. Lower thresholds produced a large number of sparse or unstable rules, while higher thresholds excluded meaningful but less frequent contributory patterns.

Rules were also filtered to be interpreted within the SHIELD causal hierarchy to ensure alignment with the multi-layer pathway (organisational > operational leadership > preconditions > acts). The filtered rules were then visualised as directed networks in which nodes represent SHIELD-NLP factors and edges represent association rules. These networks enabled qualitative exploration of how contributing factors interact within and across taxonomy layers.

ARM was conducted using expert-validated SHIELD-NLP labels, not model predictions, to ensure discovered patterns reflect the incident coding rather than classification artefacts. Formal permutation testing was not conducted in this exploratory phase. Therefore, ARM results should be interpreted as descriptive pattern discovery rather than confirmatory statistical inference.

4. Results

This section presents the performance evaluation of the proposed two-stage classification framework, which aims to automatically assign high-level categories (Stage 1) and fine-grained causal codes (Stage 2) to narrative text. In addition to quantitative classification performance, the section also reports results from association rule mining used to reveal interaction patterns between contributory factors and demonstrates how these outputs are operationally used through a prototype human-in-the-loop decision-support tool.

4.1. SHIELD-NLP construction

Applying the consolidated procedure to the original 99 SHIELD codes yielded a SHIELD-NLP taxonomy of 53 labels. All retained labels had more than 20 positive instances, reducing sparsity while preserving the multi-label structure. The overall new structure of the taxonomy is also shown in Fig. 2. Tables A1 and A2, in the appendix, showcases the full merging and removal process from SHIELD to SHIELD-NLP.

The process first merged several clusters of codes that captured the same underlying human factors, for example, perceptual codes that distinguished between auditory, kinaesthetic and other sensory channels were combined into a single category. Also, multiple communication-related codes that described unclear, incomplete, or linguistically complex exchanges were merged into one communication difficulty category. Similar consolidations occurred for organisational codes relating to personnel, training and resource limitation since these were generally overlapping issues.

Several broader psychological or cognitive constructs were also harmonised. Codes representing emotional states, confidence levels, personality tendencies and motivation were collapsed into a personal factors category. Likewise, memory-related constructs like forgetting intentions, inaccurate recall, or habitual behaviour patterns were merged into a single memory/recall factor. Visual illusions, misperception of environmental change, and misinterpretation of instruments were also integrated into a consolidated misperception category.

Lastly, a small number of codes were removed entirely where no semantically coherent merge was possible. These are typically extremely rare, conceptually unique, or structurally isolated within the taxonomy.

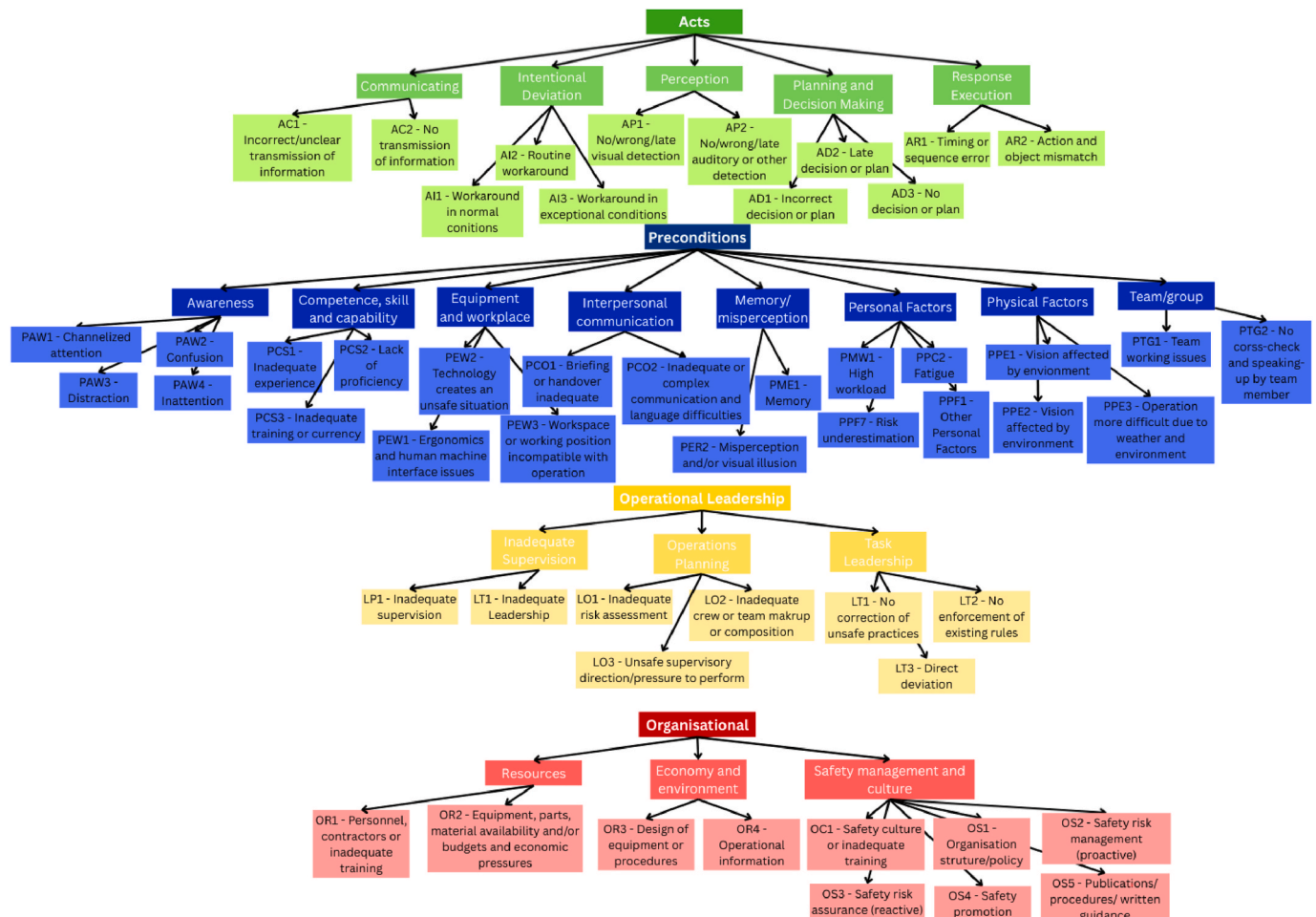


Fig. 2. SHIELD-NLP taxonomy.

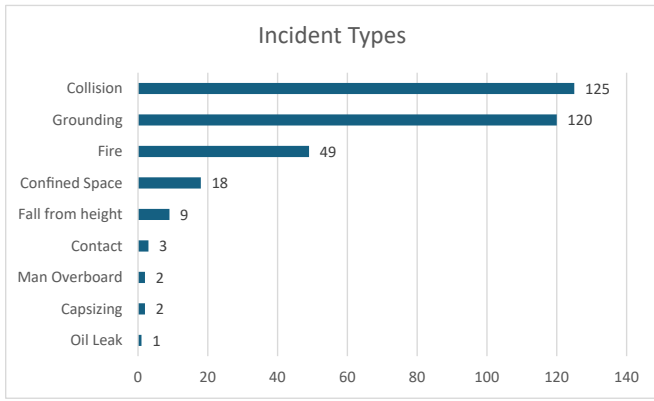


Fig. 3. Bar chart of incident types from SHIELD.

Examples include sabotage, alcohol/drug impairment, and being graphically lost. These factors could not be reliably represented through merging without distorting the taxonomy, and their low frequency made them unsuitable for machine-learning classification.

4.2. Descriptive statistics on database

The final dataset comprised 330 maritime incidents reported between 1992 and 2022. Most incidents involved no injury (68%), with 11% minor, 7% serious and 12% fatal injuries. Around half of the incidents resulted in substantial vessel damage (52%), 23% in minor damage, 14% in no damage and 10% in total loss. Collisions and groundings dominated the dataset (38% and 36% respectively), followed by fire/explosion (~15%); other incident types (e.g. confined space, falls from height, contact, capsizing, man overboard, and oil leaks occurred in much smaller numbers, shown in Fig. 3. A full table of the number of samples per code can be found in the appendix in Table A3.

Across 330 incidents, 2842 human factors were coded. In the Acts layer, the most frequent codes were ‘Incorrect decision or plan (AD1)’ (N = 251), ‘No transmission of information (AC2)’ (N = 158), and ‘No/wrong/late visual detection (AP1)’ (N = 113). Overall, decision-making failures, information omissions and perceptual errors were the dominant immediate contributors.

In the Preconditions layer, ‘Technology creates an unsafe situation (PEW2)’ was the most frequent factor (N = 87), followed by ‘Fatigue (PPC2)’ (N = 54) and ‘Inadequate experience (PCS1)’ (N = 53). A broad set of cognitive, environmental and team-related conditions were present at moderate frequencies, rather than a single dominant precondition.

Operational leadership factors were dominated by ‘Inadequate

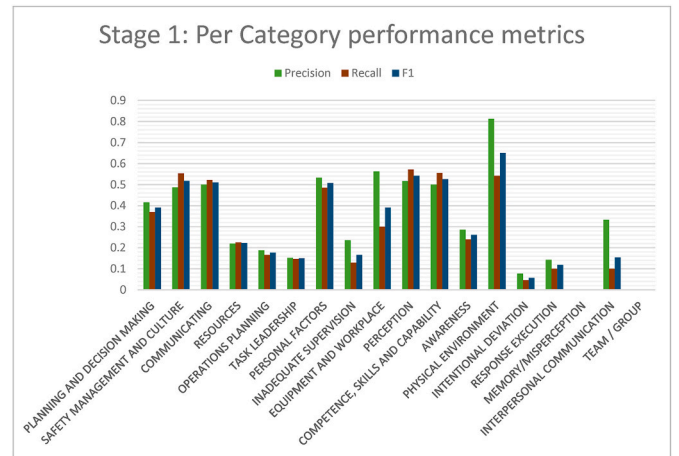


Fig. 5. Bar chart of per-category performance results.

leadership or supervision (LT1)’, the most frequent code with 123 instances, followed by ‘Inadequate risk assessment (LO1)’ with 108 instances. Furthermore, ‘No enforcement of existing rules (LT3)’ with 100 instances was also prevalent, indicating recurring weaknesses in supervisory control and procedural compliance.

Organisational factors then reflected several prominent systemic contributors with ‘Publications, procedures and written guidance (OS5)’ with 99 instances, and Equipment/parts/material availability and economic pressures (OR3) with 90 instances, highlighting the influence of procedural clarity and resource constraints on safety performance.

After cleaning, 2842 textual justifications were available. Texts were short to moderate in length, with a mean of 30.2 tokens (median 28, SD 13.4), ranging from 3 to 103 tokens. Most justifications fell between 20 and 50 tokens, shown within Fig. 4, which is well-suited to transformer-based models, as they rarely require truncation while preserving full context.

4.3. Stage 1: high-level causal category performance

For Stage 1, the model predicted 18 high-level SHIELD-NLP causal categories. As shown in Fig. 5, Table 1 and Table A4 (within the appendix). The model demonstrated strong discriminative capacity overall, with ROC-AUC values averaging 0.82, indicating substantial separability between positive and negative instances.

The highest F1-scores were obtained for Physical environment (F1 = 0.65, ROC-AUC = 0.85), Perception (F1 = 0.54, ROC-AUC = 0.96), and Competence/skills/capability (F1 = 0.53, ROC-AUC = 0.78). Several additional categories exhibited balanced and moderate performance,

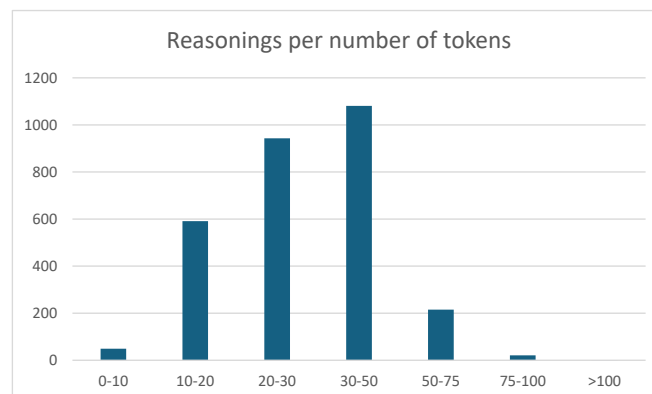


Fig. 4. Distribution of SHIELD-NLP justification text lengths measures in tokens.

Table 1

Macro and Micro summary statistics of Stage 1 model performance.

Performance Metrics	Macro	Micro
Precision	0.2623 (0.0104)	0.2475 (0.0144)
Recall	0.4872 (0.0423)	0.5114 (0.0325)
F1 score	0.3084 (0.0255)	0.3335 (0.0198)
ROC-AUC	0.8168 (0.0090)	0.8219 (0.0177)

Table 2

Macro and Micro summary statistics of Stage 1 model performance.

	Macro	Micro
Precision	0.2262 (0.0225)	0.2579 (0.0132)
Recall	0.2202 (0.0142)	0.2579 (0.0132)
F1 score	0.2115 (0.0132)	0.2579 (0.0132)
ROC-AUC	0.6546 (0.0116)	0.6682 (0.0121)

including Safety management and culture (F1 = 0.52) and Communicating (F1 = 0.51).

Lower performance occurred in categories with limited test-set support, such as intentional deviation, Response execution, Memory/misperception and Team/group. Where few or no positive predictions were made at the operating threshold. Despite this, several of these categories still achieved moderate ROC-AUC values, suggesting that class imbalance, rather than an absence of underlying signal, accounted for measurable performance.

Overall, Stage 1 results indicate that the model can reliably distinguish between most high-level human and organisational causal domains, providing a robust foundation for subsequent fine-grained classification and decision-support applications.

4.4. Per-code performance

Stage 2 evaluated the model's ability to predict 53 fine-grained SHIELD-NLP causal codes nested beneath the high-level categories. As expected for a real-world safety dataset, performance varied across codes, reflecting differences in class frequency and linguistic distinctiveness. Overall performance decreased, which is expected due to the large number of labels, substantially lower support per-code, and increased semantic overlap between finer-grained codes.

However, a substantial set of codes achieved strong classification performance, shown in Table 2 and Fig. 6, indicating that several human and/or organisational risk factors are highly learnable from narrative justifications. Shown in Table A5 (located in the appendix), several high-impact factors achieved strong or perfect F1-scores, including 'No/wrong/late visual detection (AP1)' (F1 = 1), 'No/wrong/late auditory detection (AP2)' (F1 = 1), 'Personnel/contractors/inadequate training

(OR1)' (F1 = 1), 'Inadequate risk assessment (LO1)' (F1 = 0.92), and 'Inadequate leadership or supervision (LT1)' (F1 = 0.85). Core decision-making and communication codes also performed well, e.g. 'No transmission of information (AC2)' (F1 = 0.85) and 'Incorrect decision or plan (AD1)' (F1 = 0.82). These results show that many high-impact human and organisational factors are expressed in sufficiently distinctive language to support both high precision and recall.

Among the better-performing codes, precision-recall trade-offs were evident. For example, AC2 and LT1 achieved very high recall (0.91 and 0.92, respectively), indicating that most true instances were detected at the chosen threshold, at the cost of some reduction in precision due to over-prediction. This behaviour is advantageous in safety screening and causal discovery applications, where false negatives are typically more problematic than additional candidates for expert review. Conversely, others, such as 'Vision affected by environment (PPE1)' (F1 = 0.73, precision = 1) and 'Operation more difficult due to weather (PPE6)' (F1 = 0.67, precision = 1), exhibited very high precision but more conservative recall, favouring high-confidence automated flagging.

A small group of rare, abstract codes (e.g. OS1, LP1, PPF1, PAW2, AI2) returned F1 = 0 despite moderate ROC-AUC values. These categories share two characteristics: they have very low test-set support and represent abstract semantic constructs often described only indirectly in the text. Overall, this pattern supports the conclusion that data scarcity, rather than model capacity, is the primary limitation for these codes.

4.5. Model performance comparisons

To quantify the contribution of domain adaptation and hybrid lexical-contextual representations, a comparison was conducted for the four feature representations: TF-IDF only, generic BERT, SeaSafeBERT, and the hybrid representation proposed. All models used the same multinomial logistic regression classifier and were evaluated using 5-fold stratified cross-validation, reporting mean and variability across folds.

Across folds, contextual transformer embeddings substantially outperformed TF-IDF, as shown in Table 3. Among transformer variants, the hybrid representation model achieved the highest performance. Although the differences between the models were modest, suggesting that domain adaptation provides incremental gains under the current corpus size.

4.6. Association rule mining network

Association rule mining (ARM) was used to identify patterns of co-occurring contributory factors within the SHIELD taxonomy. Rules were first extracted from the full dataset and then repeated for major accident types to understand how interaction patterns may vary by

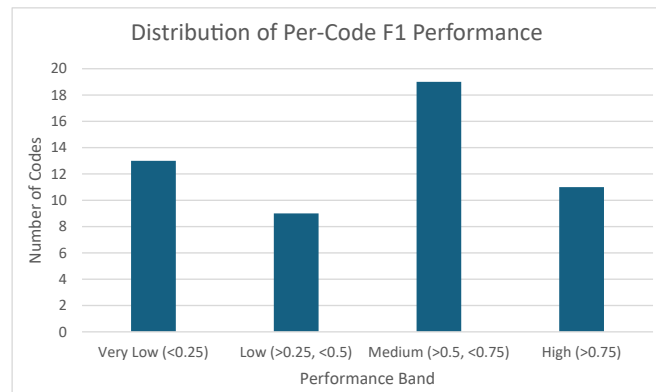


Fig. 6. Bar chart showing the number of codes in each performance band with F1 score.

Table 3
Model performance comparison across different feature representation methods.

Model	Macro-F1	Micro-F1	Macro-AUC	Micro-AUC
TF-IDF	0.0460 (± 0.0506)	0.0605 (± 0.0751)	0.7646 (± 0.0181)	0.6917 (± 0.0520)
Generic BERT	0.2943 (± 0.0236)	0.3147 (± 0.0173)	0.8070 (± 0.0100)	0.8150 (± 0.0151)
SeaSafeBERT	0.2958 (± 0.0114)	0.3207 (± 0.0096)	0.8080 (± 0.0080)	0.8167 (± 0.0035)
Hybrid Model	0.3084 (± 0.0254)	0.3335 (± 0.0198)	0.8168 (± 0.0090)	0.8219 (± 0.0176)

Causal Association Rule Graph (All Accidents)

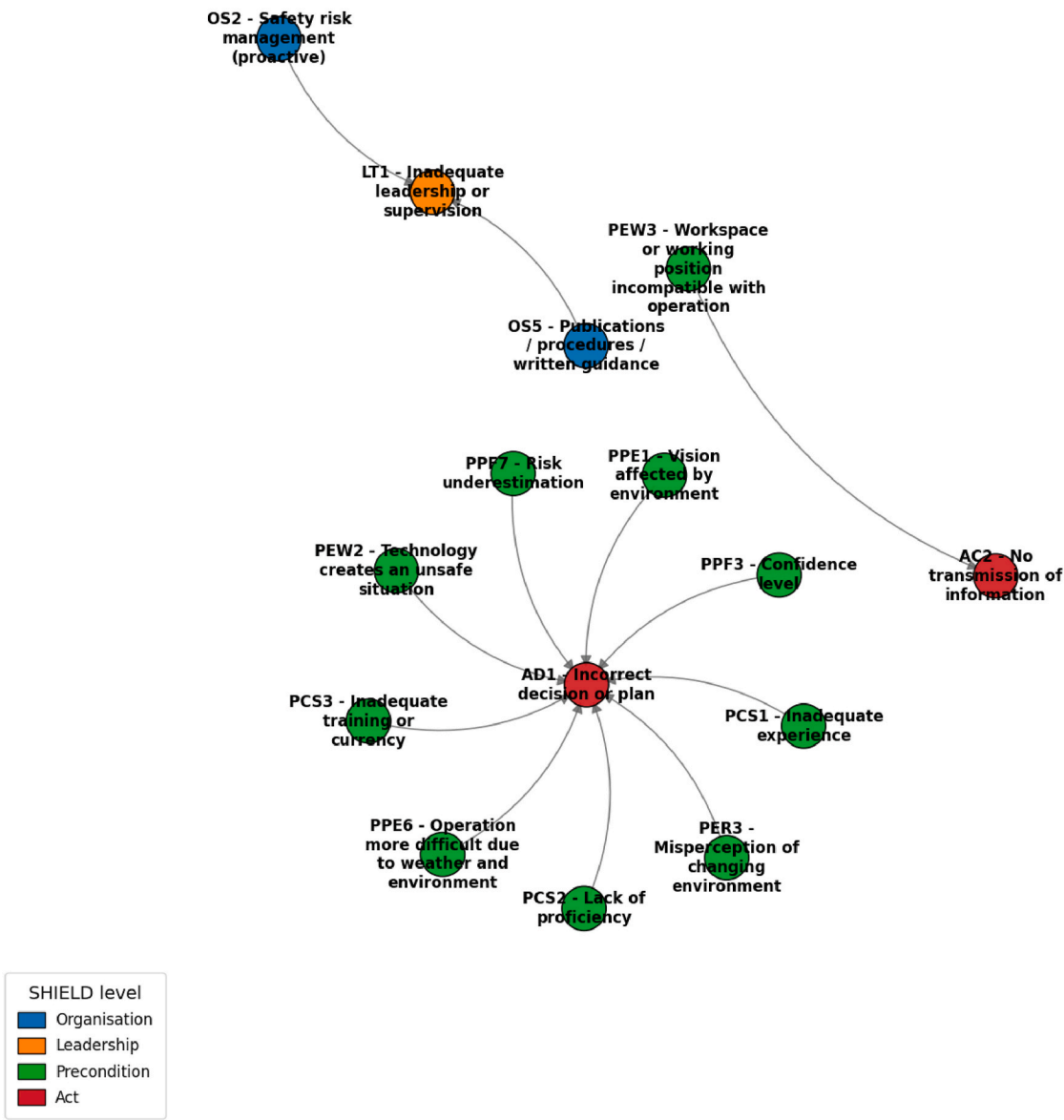


Fig. 7. ARM network of contributory factors (full dataset).

Table 4

Full data association rule mining results.

antecedents	consequents	support	confidence	lift
['PEW3 - Workspace or working position incompatible with operation']	['AC2 - No transmission of information']	0.051	0.613	1.763
['PPE1 - Vision affected by environment']	['AD1 - Incorrect decision or plan']	0.070	0.619	1.196
['PPE6 - Operation more difficult due to weather and environment']	['AD1 - Incorrect decision or plan']	0.065	0.615	1.189
['PEW2 - Technology creates an unsafe situation']	['AD1 - Incorrect decision or plan']	0.089	0.600	1.159
['PPF3 - Confidence level']	['AD1 - Incorrect decision or plan']	0.054	0.800	1.546
['PPF7 - Risk underestimation']	['AD1 - Incorrect decision or plan']	0.078	0.763	1.475
['PCS1 - Inadequate experience']	['AD1 - Incorrect decision or plan']	0.111	0.672	1.299
['PCS2 - Lack of proficiency']	['AD1 - Incorrect decision or plan']	0.097	0.706	1.364
['PCS3 - Inadequate training or currency']	['AD1 - Incorrect decision or plan']	0.057	0.724	1.399
['PER3 - Misperception of changing environment']	['AD1 - Incorrect decision or plan']	0.054	0.870	1.680
['OS5 - Publications/procedures/written guidance', 'OS2 - Safety risk management (proactive)']	['LT1 - Inadequate leadership or supervision']	0.051	0.679	2.289

context.

4.6.1. Full data

ARM generated 11 meaningful associations, shown in Fig. 7 and Table 4, indicating strong interdependence between perceptual, cognitive, organisational, and environmental factors. A consistent theme concerned the influence of workspace and equipment design on communication quality: workspace incompatible with operation, resulting in no transmission of information (confidence = 0.61), suggesting that ergonomic constraints frequently impair information flow.

Decision-making failures were another major cluster. Factors such as misperception of the environment (confidence = 0.87), risk underestimation (confidence = 0.80), and inadequate training (confidence = 0.72) strongly predicted incorrect decision or plan, indicating that situational awareness deficits and competence gaps are central drivers of human error.

Organisational precursors also emerged prominently. Weak procedures combined with poor safety risk management were commonly associated with inadequate leadership or supervision, highlighting how systemic governance issues cascade into frontline supervisory failures.

4.6.2. Collision accidents

Collision cases show pronounced perceptual-environmental interactions, shown in Fig. 8 and Table 5. Visual detection failures were frequently linked to unsafe technology, poor workspace ergonomics, and inadequate communication equipment. Several rules reached confidence values of 1.0, confirming that degraded perceptual conditions almost invariably co-occur with equipment or workspace shortcomings.

Cognitive overload and attentional narrowing were also repeatedly associated with no transmission of information, indicating that impaired perception not only reduces detection but also disrupts shared situational awareness.

Decision-making errors were strongly linked to risk underestimation (confidence = 0.93) and competence-related factors, underscoring the role of cognitive bias and experience gaps in collision scenarios.

At the organisational level, deficiencies in policy and procedural clarity were consistently associated with failures in rule enforcement and supervision, reflecting structural weaknesses contributing to navigational risk.

4.6.3. Grounding

Fig. 9 and Table 6 shows that grounding incidents were characterised by cognitive limitations and environmental stressors. Confusion, workload, and risk underestimation frequently predicted communication failures, while adverse weather and unsafe technology were repeatedly associated with incorrect decision-making.

Competence-related factors (PCS1-PCS3) showed some of the strongest associations in the dataset, with several rules exhibiting confidence values of 1, highlighting the critical influence of proficiency and training on grounding risk.

Organisational shortcomings in safety promotion and risk management were also linked to poor supervisory performance, demonstrating how weak governance propagates into operational decision-making.

4.6.4. Fire

Fire-related accidents exhibited stronger environmental and cognitive patterns than other categories, as shown in Fig. 10 and Table 7. To enhance interpretability, only representative rules from each dominant causal pathway were retained in the summarisation table. Memory failures reliably preceded timing errors, with combined antecedents producing high-confidence relationships (0.75).

Severe weather, heat stress, and other environmental stressors were associated with both perceptual failures and incorrect decisions, indicating that physiological overload can rapidly degrade performance in high-risk fire contexts.

Organisational factors such as inadequate risk assessment often appeared upstream of these failures, suggesting that poor planning exposes crews to conditions that exceed human performance limits.

4.6.5. Occupational

Occupational accidents showed distinctive behavioural patterns, as shown in Fig. 11 and Table 8. Negative habits emerged as one of the strongest predictors of both communication failures and decision-making errors, with several rules reaching perfect confidence and very high lift values.

Competence deficits and risk underestimation also played major roles, but unlike navigational accidents, occupational incidents displayed a closer link between habitual behaviour, procedural drift, and operational shortcuts.

Organisational weaknesses in rule enforcement and risk assessment were again associated with inadequate briefing or handover, reinforcing the systemic nature of workplace accidents.

5. Discussion

5.1. Overview of findings

5.1.1. Classification performance

Overall, the findings indicate that an NLP pipeline combining domain-adapted language representations (SeaSafeBERT), hierarchical logistic regression with calibration, and human-in-the-loop decision logic, demonstrates meaningful discriminative capacity for many contributory factors from narrative reports and surface meaningful

Causal Association Rule Graph (Collision Accidents)

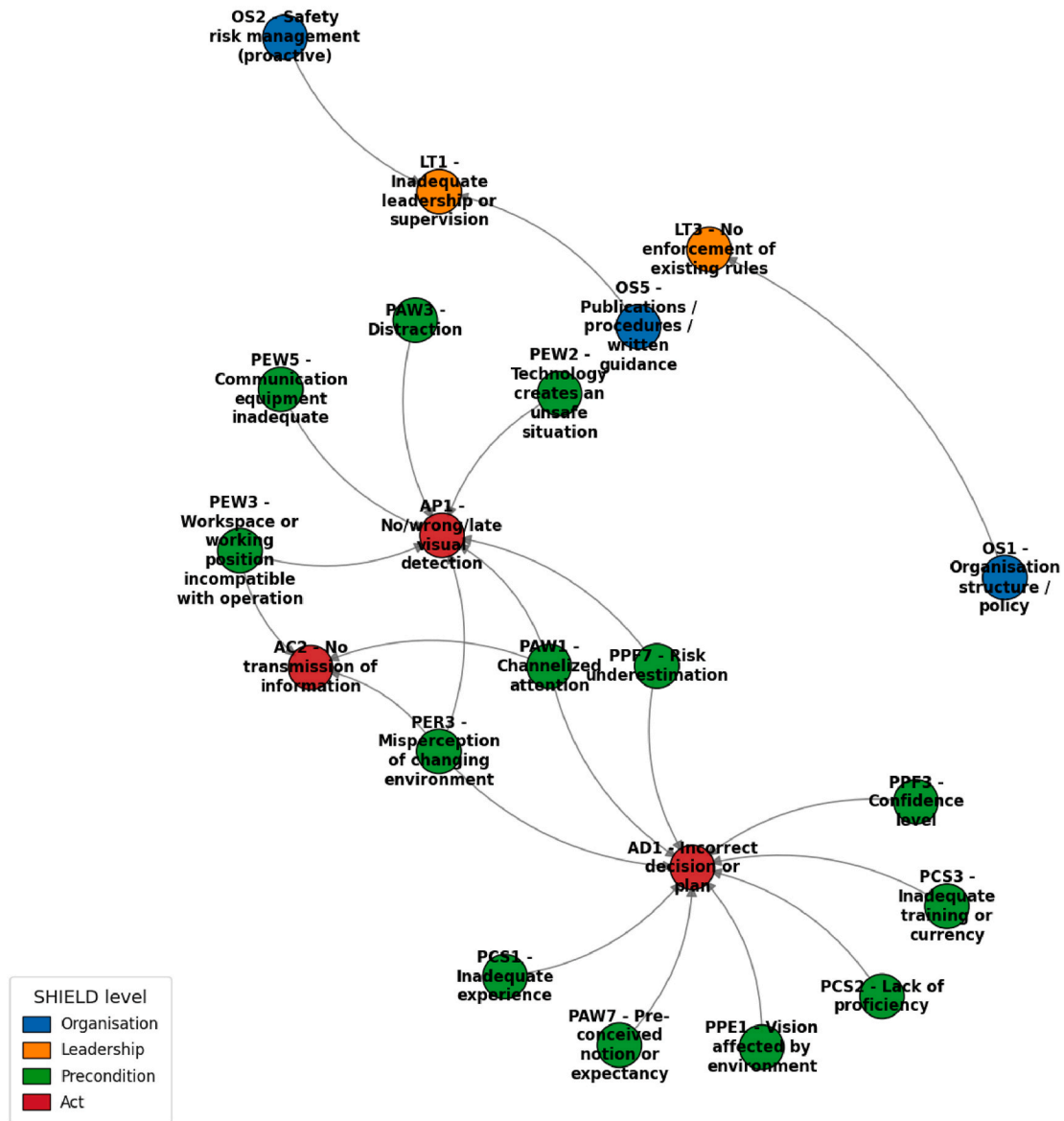


Fig. 8. ARM network of contributory factors for collision incidents.

interaction patterns between them.

The stage 1 modelling results showed that high-level categories tied to concrete scene descriptors (e.g. Physical Environment, Perception, Competence/Capability) are detected with comparatively strong F1 and ROC-AUC, whereas abstract or sparsely represented categories (e.g. International Deviation, Response Execution) remain challenging.

At the code level, several factors exhibit operationally useful accuracy, notably no wrong/late visual detection (AP1), no transmission of information (AC2), incorrect decision or plan (AD1), briefing or

handover inadequate (PCO1), inadequate leadership or supervision (LT1), inadequate risk assessment (LO1), crew composition (LO2) and personnel/training resources (OR1) achieved very high scores. These codes achieved strong precision-recall balance (many with $F1 \geq 0.80$), suggesting that they may be suitable candidates for high-confidence automated flagging within a supervised, human-in-the-loop workflow.

The precision-recall trade-off was evident between other strong performers, suggesting complementary deployment modes since high recall models can drive broad triage and discovery, while high precision

Table 5
Collision incidents association rule mining results.

antecedents	consequents	support	confidence	lift
[PEW2 - Technology creates an unsafe situation]	[AP1 - No/wrong/late visual detection]	0.10	0.68	1.46
[PEW3 - Workspace or working position incompatible with operation]	[AP1 - No/wrong/late visual detection]	0.07	1.00	2.14
[PEW5 - Communication equipment inadequate]	[AP1 - No/wrong/late visual detection]	0.06	0.88	1.87
[PAW1 - Channelized attention]	[AP1 - No/wrong/late visual detection]	0.06	0.64	1.36
[PAW3 - Distraction]	[AP1 - No/wrong/late visual detection]	0.11	0.61	1.30
[PPF7 - Risk underestimation]	[AP1 - No/wrong/late visual detection]	0.07	0.60	1.28
[PER3 - Misperception of changing environment]	[AP1 - No/wrong/late visual detection]	0.07	0.64	1.37
[PEW3 - Workspace or working position incompatible with operation]	[AC2 - No transmission of information]	0.06	0.78	1.79
[PAW1 - Channelized attention]	[AC2 - No transmission of information]	0.06	0.73	1.67
[PER3 - Misperception of changing environment]	[AC2 - No transmission of information]	0.07	0.64	1.48
[PPE1 - Vision affected by environment]	[AD1 - Incorrect decision or plan]	0.14	0.61	0.94
[PAW1 - Channelized attention]	[AD1 - Incorrect decision or plan]	0.06	0.64	0.99
[PAW7 - Pre-conceived notion or expectancy]	[AD1 - Incorrect decision or plan]	0.06	0.89	1.38
[PPF3 - Confidence level]	[AD1 - Incorrect decision or plan]	0.07	0.69	1.07
[PPF7 - Risk underestimation]	[AD1 - Incorrect decision or plan]	0.11	0.93	1.45
[PCS1 - Inadequate experience]	[AD1 - Incorrect decision or plan]	0.18	0.69	1.07
[PCS2 - Lack of proficiency]	[AD1 - Incorrect decision or plan]	0.13	0.80	1.24
[PCS3 - Inadequate training or currency]	[AD1 - Incorrect decision or plan]	0.06	0.67	1.03
[PER3 - Misperception of changing environment]	[AD1 - Incorrect decision or plan]	0.10	0.93	1.44
[OS1 - Organisation structure/policy]	[LT3 - No enforcement of existing rules]	0.08	0.77	2.58
[PEW3 - Workspace or working position incompatible with operation]	[AP1 - No/wrong/late visual detection', 'AC2 - No transmission of information]	0.06	0.78	3.21
[PPF7 - Risk underestimation]	[AP1 - No/wrong/late visual detection', 'AD1 - Incorrect decision or plan']	0.07	0.60	2.07
[PER3 - Misperception of changing environment]	[AP1 - No/wrong/late visual detection', 'AD1 - Incorrect decision or plan']	0.07	0.64	2.21
[PER3 - Misperception of changing environment]	[AC2 - No transmission of information', 'AD1 -	0.07	0.64	2.10

Table 5 (continued)

antecedents	consequents	support	confidence	lift
	Incorrect decision or plan']			
[PPE1 - Vision affected by environment', 'PCS1 - Inadequate experience']	[AD1 - Incorrect decision or plan']	0.06	0.67	1.03
[OS2 - Safety risk management (proactive)', 'OS5 - Publications/procedures/written guidance']	[LT1 - Inadequate leadership or supervision']	0.06	0.78	2.19

models support low-volume, high-confidence automation. These help support a mixed workflow where analysts receive both auto-coded factors and calibrated suggestions.

5.1.2. Accident patterns identified through ARM

The ARM analyses added explanatory context by highlighting how factors cluster within and across accident types. One key similarity between accidents lies in the prominence of cognitive and attentional precursors, such as risk underestimation, inattention, and memory-related errors, consistently leading to incorrect decision-making. Across all accident types, the confidence of rules linking these factors to decision failures is high, indicating a consistently observed co-occurrence pattern within the analysed dataset. This pattern is consistent with prior research emphasising the importance of cognitive and organisational factors in maritime incidents.

Despite this commonality, the mechanisms by which cognitive limitation manifest different by accident type. In a collision-scenarios, cognitive failures are often intertwined with perceptual issues, particularly failures in visual detection. This is distinct from grounding accidents, where the emphasis is more on decision-making under complex or uncertain environmental conditions, such as poor weather or technological confusion. Fire-related incidents, on the other hand, are unique in showing comparatively stronger multi-antecedent associations involving memory-based antecedents and physiological stress, often resulting in both timing and planning failures simultaneously. This dual-error phenomenon was not as prominent in other accident types.

Another important similarity is the recurrent role of organisational factors in escalating risk. In all four types, rules involving inadequate leadership, safety management, and lack of enforcement or structure appear frequently as antecedents to high-level consequences like inadequate risk assessment or inadequate briefing or handover. However, the way these organisational deficiencies play out varies. In collisions and groundings, organisational issues tend to result in supervisory or procedural lapses, often indirectly contributing to perceptual or planning errors. In fire and occupational accidents, these issues more directly result in breakdowns of communication and informal adaptations, such as workarounds, which suggest a behavioural response to systemic inadequacies.

A notable difference is the degree of complexity and convergence present in the rules. Fire and occupational accidents display the highest prevalence of multi-antecedent rules, often with perfect confidence and elevated lift values. This suggests a latent risk accumulation mode, where multiple collision and grounding accidents tend to be explained by fewer, more direct associations, often involving one or two key antecedents leading to a single outcome. This could reflect the more situational and dynamic nature of these accidents, where failure occurs more rapidly and is less dependent on long-term latent conditions.

Another key difference lies in the role of environmental stressors. These were especially prominent in fire and grounding incidents, contributing to both cognitive and perceptual failures. In contrast, such environmental factors were almost absent in occupational accidents,

Causal Association Rule Graph (Grounding Accidents)

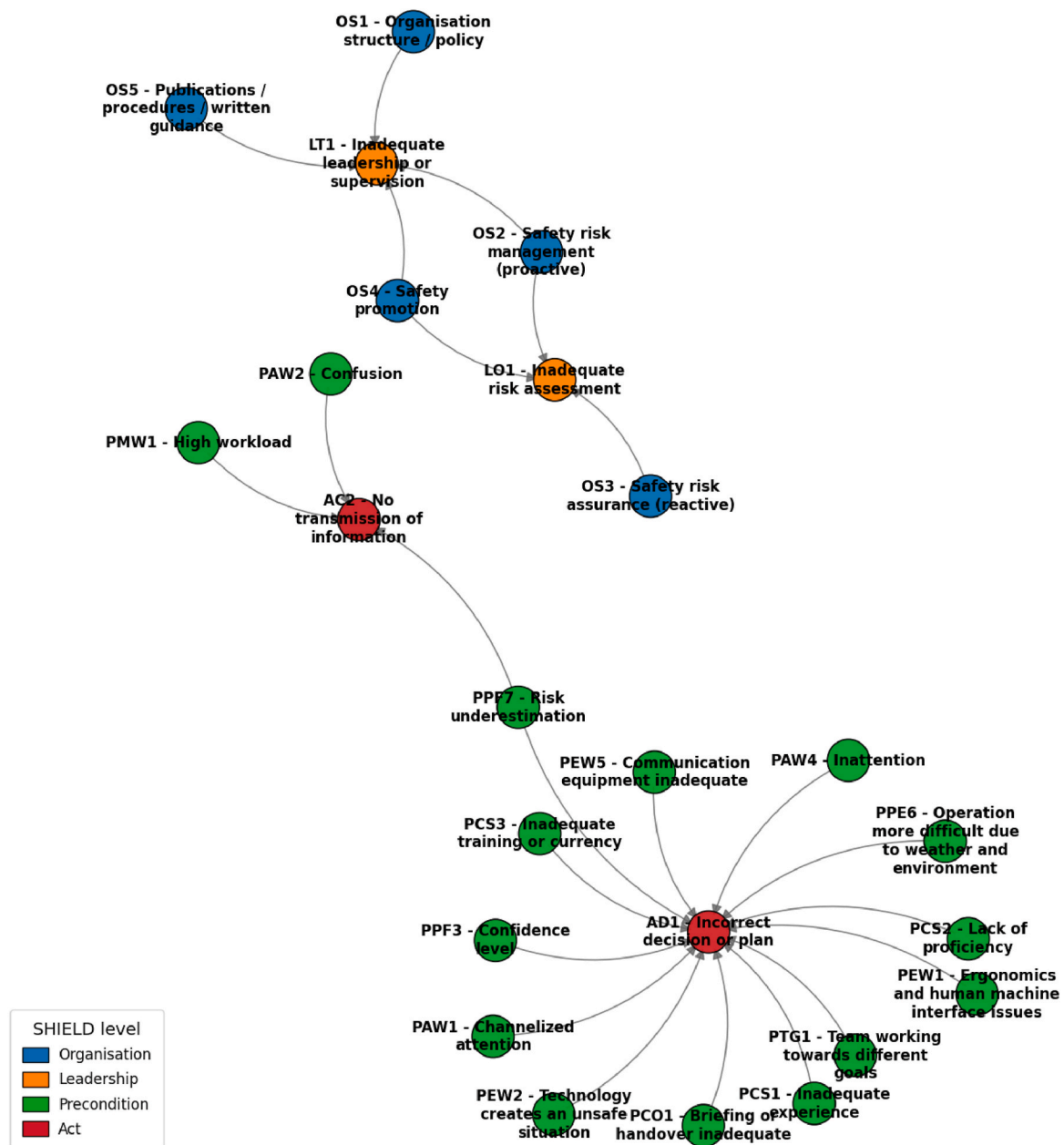


Fig. 9. ARM network of contributory factors for grounding incidents.

where behavioural habits and procedural breakdowns were the dominant themes. Lastly, only collision and fire accidents displayed rules with multiple consequences, suggesting divergent causality where single antecedents lead to both decision and perceptual failures. This suggests that certain unsafe conditions can initiate a cascade of failures affecting multiple operational layers. This was evident in grounding and occupational incidents, where causal pathways were more linear and targeted.

Overall, all accident types share a foundational vulnerability rooted

in cognitive limitations and organisational deficiencies, their expressions differ based on context. Collisions are characterised by perceptual breakdowns, groundings by environmental complexity and inexperience, fires by cognitive overload under stress and occupational accidents by entrenched behavioural patterns and weak procedural oversight. These insights suggest that while certain interventions may benefit all domains, each accident type also requires tailored strategies that reflect its unique risk dynamics.

Table 6
Grounding incidents association rule mining results.

antecedents	consequents	support	confidence	lift
['PAW2 - Confusion']	['AC2 - No transmission of information']	0.07	0.67	1.85
['PMW1 - High workload']	['AC2 - No transmission of information']	0.07	0.64	1.78
['PPF7 - Risk underestimation']	['AC2 - No transmission of information']	0.07	0.67	1.85
['PPE6 - Operation more difficult due to weather and environment']	['AD1 - Incorrect decision or plan']	0.12	0.65	1.06
['PEW1 - Ergonomics and human machine interface issues']	['AD1 - Incorrect decision or plan']	0.10	0.60	0.98
['PEW2 - Technology creates an unsafe situation']	['AD1 - Incorrect decision or plan']	0.08	0.67	1.08
['PEW5 - Communication equipment inadequate']	['AD1 - Incorrect decision or plan']	0.06	0.70	1.14
['PCO1 - Briefing or handover inadequate']	['AD1 - Incorrect decision or plan']	0.07	0.75	1.22
['PTG1 - Team working towards different goals']	['AD1 - Incorrect decision or plan']	0.07	0.75	1.22
['PAW1 - Channelized attention']	['AD1 - Incorrect decision or plan']	0.07	0.82	1.33
['PAW4 - Inattention']	['AD1 - Incorrect decision or plan']	0.07	0.60	0.98
['PPF3 - Confidence level']	['AD1 - Incorrect decision or plan']	0.08	0.91	1.48
['PPF7 - Risk underestimation']	['AD1 - Incorrect decision or plan']	0.08	0.83	1.36
['PCS1 - Inadequate experience']	['AD1 - Incorrect decision or plan']	0.11	0.67	1.08
['PCS2 - Lack of proficiency']	['AD1 - Incorrect decision or plan']	0.14	0.68	1.11
['PCS3 - Inadequate training or currency']	['AD1 - Incorrect decision or plan']	0.07	0.82	1.33
['OS4 - Safety promotion']	['LO1 - Inadequate risk assessment']	0.09	0.65	1.75
['PCS2 - Lack of proficiency', 'PCS3 - Inadequate training or currency']	['AD1 - Incorrect decision or plan']	0.06	1.00	1.63
['OS4 - Safety promotion', 'OS2 - Safety risk management (proactive)']	['LO1 - Inadequate risk assessment']	0.07	0.73	1.97
['OS3 - Safety risk assurance (reactive)', 'OS4 - Safety promotion']	['LO1 - Inadequate risk assessment']	0.06	0.88	2.37
['OS1 - Organisation structure/policy', 'OS5 - Publications/procedures/written guidance']	['LT1 - Inadequate leadership or supervision']	0.07	0.73	1.74
['OS4 - Safety promotion', 'OS2 - Safety risk management (proactive)']	['LT1 - Inadequate leadership or supervision']	0.07	0.73	1.74
['OS5 - Publications/procedures/written guidance', 'OS2 - Safety risk management (proactive)']	['LT1 - Inadequate leadership or supervision']	0.08	0.63	1.50
['OS4 - Safety promotion', 'OS5 - Publications/procedures/written guidance']	['LT1 - Inadequate leadership or supervision']	0.06	0.78	1.86
['OS4 - Safety promotion', 'OS5 - Publications/procedures/written guidance', 'OS2 - Safety risk management (proactive)']	['LT1 - Inadequate leadership or supervision']	0.06	0.88	2.09

5.2. Methodological implications

Beyond classification accuracy, the methodology provides a structured and potentially deployable framework, a suitable pathway for AI-assisted investigation. The SHIELD-NLP label set balanced causal factor complexity with learnability, preserving the intent of SHIELD while ensuring each class has sufficient support for stable training. The hybrid representation (SeaSafeBERT embeddings with TF-IDF) captures maritime semantics and the high-precision operational cues. The painting consistently improved precision for codes whose evidence is expressed through distinctive phrasing.

The hierarchical, calibrated OVR elastic net framework emphasises interpretability and control. Analysts can inspect coefficients and through probability calibration using the auto/suggest method can understand the extent to which the model is sure of the classification. Future work on embedding these into an interface tool turns the model into a learning system where submission of classifications is re-added to the model's database. Coupling the modelling process with rule-based coding also helped with very low sample sizes, allowing us not remove them.

ARM then adds explanatory context of how these factors co-occur and breaking down to incident types allows us to differentiate and create actionable hypotheses for design, training, and governance interventions. Together these methods demonstrate the potential to support more consistent, and scalable learning from accidents at sea, subject to continued validation and oversight.

The deployment of AI-assisted classification in safety-critical domains requires careful governance. Automated outputs should be interpreted as probabilistic suggestions rather than authoritative determinations. Over-reliance on automated coding risks reinforcing historical attribution biases embedded in past investigations. For this reason, the proposed system is designed as a human-in-the-loop decision-support tool, where final coding authority remains with trained analysts.

5.3. Limitations

Despite these encouraging results, performance was uneven across the taxonomy. Several fine-grained or abstract categories exhibited low or near-zero F1-scores, despite moderate ROC-AUC values. This indicates that while the model may capture the underlying signal, class imbalance and limited linguistic distinctiveness restrict reliable deployment for those factors. The system therefore performs robustly for well-represented and concretely described codes, but remains limited for rare or conceptually diffuse constructs.

The dataset also comprises English-language maritime accident investigation reports drawn from a limited number of authorities. While these organisations produce higher-quality reports, their narrative style, terminology and investigative emphasis may differ from those of other jurisdictions. Consequently, the linguistic patterns learned by the model may partially reflect reporting conventions rather than universally stable causal structures. Generalisation to non-English reports, alternative regulatory contexts, or differently structured investigation formats cannot be assumed without further validation.

Additionally, the transformation of the original 99 SHIELD codes into the 53-label SHIELD-NLP taxonomy required merging semantically similar codes and excluding extremely low-frequency categories. While this step was necessary to enable stable supervised learning, it may have reduced conceptual granularity and blurred distinctions between closely related human factors.

The modelling pipeline performs data splitting at the sentence level rather than the report level. This reflects the intended operational scenario in which individual causal statements are classified independently. However, this approach introduces a potential risk that sentences from the same report may appear in both training and evaluation partitions. Although no raw labels were shared across splits and

Causal Association Rule Graph (Fire Accidents)

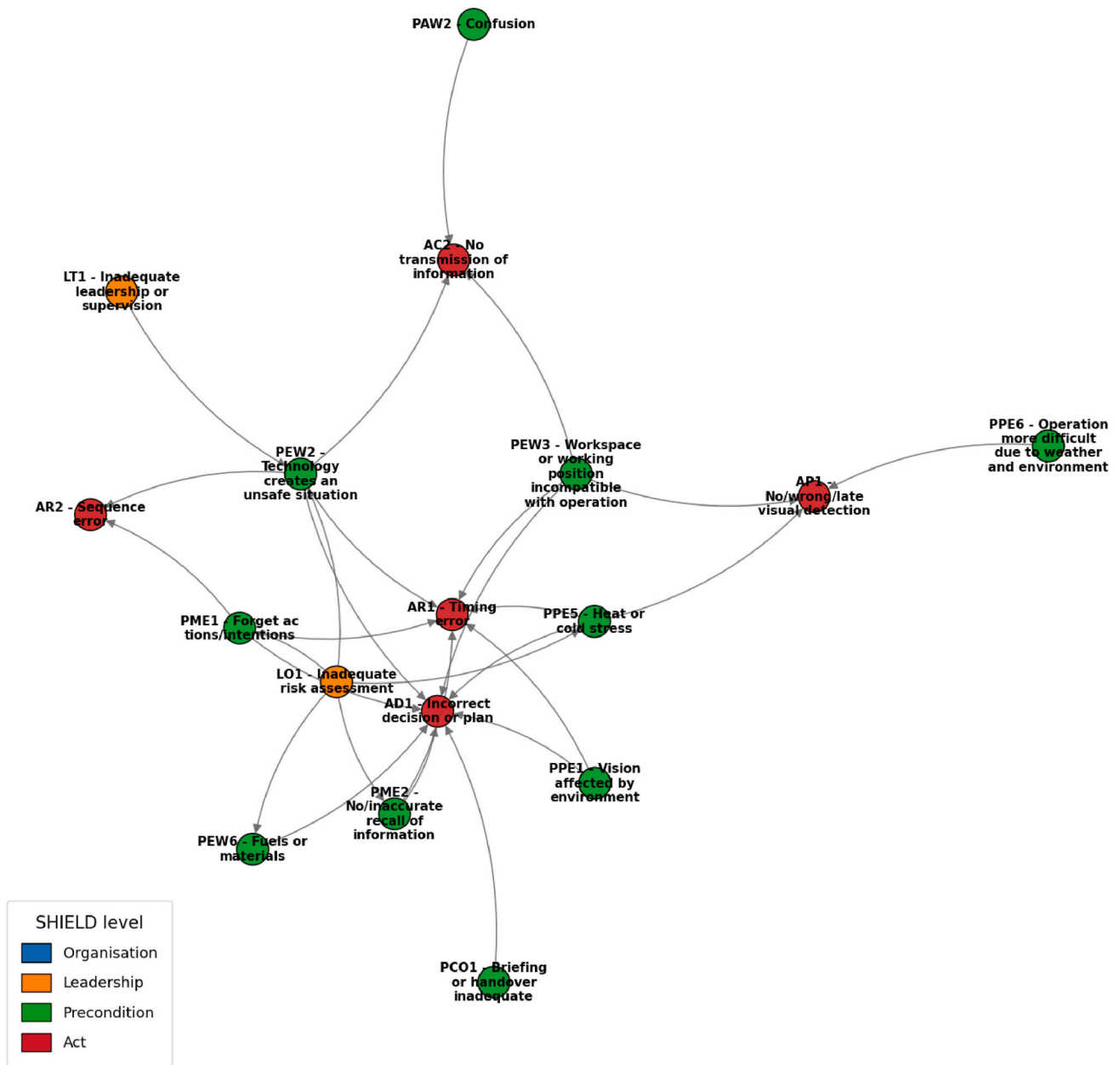


Fig. 10. ARM network of contributory factors in fire incidents.

resampling, calibration, and threshold tuning were strictly confined to training folds.

Similarly, to address class imbalance, the modelling pipeline incorporates SMOTE-based oversampling and class weighting. While these techniques improve minority-class recall, they introduce synthetic training examples and alter effective class distributions within the training folds. Synthetic samples may amplify noise in rare categories or exaggerate local decision boundaries. Although resampling was confined strictly to training subsets to prevent data leakage,

performance estimates may still reflect artefacts of the resampling strategy rather than natural real-world prevalence.

5.4. Future work

The findings of this study highlight several promising directions for future research that can build upon and extend the methodological and practical contributions. While the system demonstrated encouraging performance in several categories for supporting large-scale human-

Table 7
Fire incidents association rule mining results.

antecedents	consequents	support	confidence	lift
['PME1 - Forget actions/ intentions']	['AR1 - Timing error']	0.08	0.67	4.17
['PME1 - Forget actions/ intentions']	['AD1 - Incorrect decision or plan']	0.1	0.83	1.89
['PME2 - No/inaccurate recall of information']	['AD1 - Incorrect decision or plan']	0.1	0.71	1.62
['LO1 - Inadequate risk assessment']	['PPE5 - Heat or cold stress']	0.08	0.8	3.33
['LT1 - Inadequate leadership or supervision']	['PEW2 - Technology creates an unsafe situation']	0.08	0.67	1.85
['PPE5 - Heat or cold stress', 'PEW3 - Workspace or working position incompatible with operation']	['AR1 - Timing error']	0.08	0.8	5
['PPE5 - Heat or cold stress', 'PME2 - No/inaccurate recall of information']	['AR1 - Timing error']	0.08	0.67	4.17
['PME2 - No/inaccurate recall of information', 'PEW3 - Workspace or working position incompatible with operation']	['AR1 - Timing error']	0.08	0.8	5
['PEW2 - Technology creates an unsafe situation', 'PEW3 - Workspace or working position incompatible with operation']	['AC2 - No transmission of information']	0.1	0.63	1.36
['PEW2 - Technology creates an unsafe situation', 'PPE5 - Heat or cold stress']	['AD1 - Incorrect decision or plan']	0.12	0.6	1.36
['PPE5 - Heat or cold stress', 'PEW3 - Workspace or working position incompatible with operation']	['AD1 - Incorrect decision or plan']	0.08	0.8	1.82
['PPE5 - Heat or cold stress', 'PME2 - No/inaccurate recall of information']	['AD1 - Incorrect decision or plan']	0.1	0.83	1.89
['PEW2 - Technology creates an unsafe situation', 'PME1 - Forget actions/ intentions']	['AD1 - Incorrect decision or plan']	0.1	1	2.27
['PEW2 - Technology creates an unsafe situation', 'PME2 - No/ inaccurate recall of information']	['AD1 - Incorrect decision or plan']	0.1	0.83	1.89
['PME1 - Forget actions/ intentions', 'PEW6 - Fuels or materials']	['AD1 - Incorrect decision or plan']	0.08	1	2.27
['PPE5 - Heat or cold stress', 'PME2 - No/inaccurate recall of information', 'PEW3 - Workspace or working position incompatible with operation']	['AR1 - Timing error']	0.08	1	6.25
['PEW2 - Technology creates an unsafe situation', 'PME2 - No/ inaccurate recall of information', 'PPE5 - Heat or cold stress']	['AD1 - Incorrect decision or plan']	0.1	1	2.27
['PEW2 - Technology creates an unsafe situation', 'PME1 - Forget actions/ intentions', 'PEW6 - Fuels or materials']	['AD1 - Incorrect decision or plan']	0.08	1	2.27

factors analysis, further work is required to enhance generalisability, robustness, and operational integration.

Expanding data breadth and linguistic coverage is a key priority. Several low-frequency SHIELD codes exhibited limited predictive performance, reflecting the inherent imbalance of maritime safety datasets. Future studies should incorporate larger and more diverse corpora and extend the approach to multilingual or translated reports.

As an area for future work, integrating these models into a dedicated user interface could further enhance human-in-the-loop effectiveness. Such an interface could support collaborative features, including cross-analysis comparison and disagreement tracking, to help identify systematic ambiguities in taxonomy interpretation. Another critical next step is the evaluation of the tool in operational settings. Controlled field studies could assess reductions in analysis time, improvements in consistency across investigators, and the influence of AI-supported coding on the quality of corrective actions and safety recommendations.

Beyond model and interface improvements, future work should also address the upstream structuring of incident reporting itself. Variability in report completeness, narrative detail, and causal explicitness currently constrains automated analysis. Developing reporting guidelines or semi-structured templates that encourage clearer articulation of causal reasoning, contextual conditions, and organisational factors would improve both human interpretability and machine-readability. Integrating such structured reporting elements directly into investigation workflows could significantly enhance the quality and consistency of data available for automated analysis.

6. Conclusion

This study demonstrated the feasibility and value of applying AI-driven text analytics to automate the extraction of human and organisational factors from maritime incident reports using the SHIELD taxonomy. By developing a domain-specific language model, a hybrid technical classification pipeline, and an operational prototype tool, the research shows that many SHIELD-NLP codes can be identified with high reliability and consistency. The integration of association rule mining further revealed meaningful patterns linking perceptual, cognitive, environmental, and organisational contributors across different incident types, offering new insights into systemic accident causation. While some challenges remain, particularly for rare of abstract codes, the results indicate that automated analysis can substantially enhance the speed, scalability, and quality of safety learning. Overall, this work provides a foundation for more proactive, data-driven maritime safety management and establishes clear pathways for future expansion across larger and more diverse datasets.

CRedit authorship contribution statement

Hollie Black: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Rafet Kurt:** Writing – review & editing, Supervision, Conceptualization. **Osman Turan:** Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Causal Association Rule Graph (Occupational Accidents)

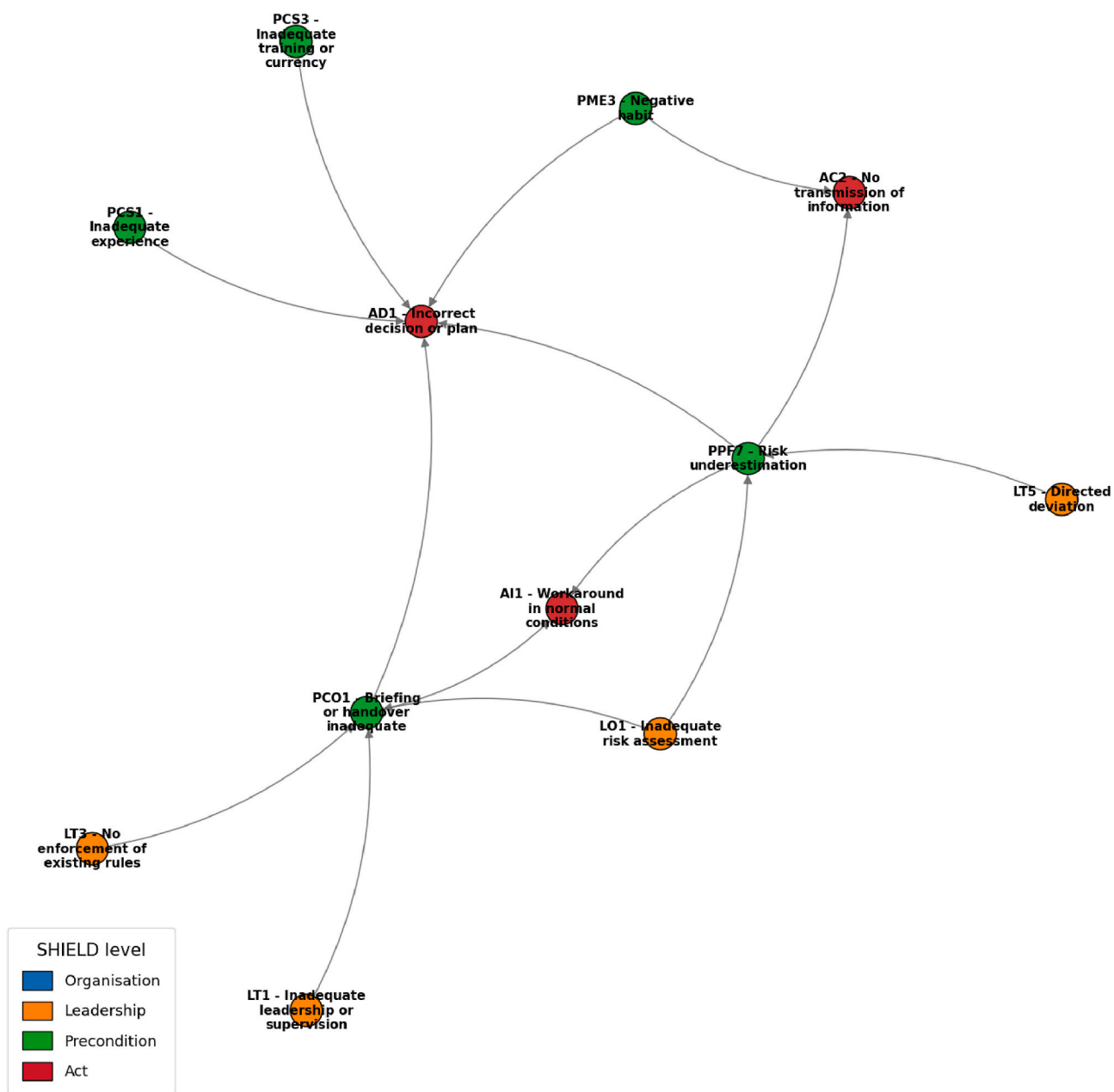


Fig. 11. ARM network of contributory factors in occupational incidents.

Table 8
Occupational incidents association rule mining results.

antecedents	consequents	support	confidence	lift
['PEW3 - Workspace or working position incompatible with operation']	['AC2 - No transmission of information']	0.05	0.61	1.76
['PPE1 - Vision affected by environment']	['AD1 - Incorrect decision or plan']	0.07	0.62	1.20
['PPE6 - Operation more difficult due to weather and environment']	['AD1 - Incorrect decision or plan']	0.06	0.62	1.19
['PEW2 - Technology creates an unsafe situation']	['AD1 - Incorrect decision or plan']	0.09	0.60	1.16
['PPF3 - Confidence level']	['AD1 - Incorrect decision or plan']	0.05	0.80	1.55
['PPF7 - Risk underestimation']	['AD1 - Incorrect decision or plan']	0.08	0.76	1.47
['PCS1 - Inadequate experience']	['AD1 - Incorrect decision or plan']	0.11	0.67	1.30
['PCS2 - Lack of proficiency']	['AD1 - Incorrect decision or plan']	0.10	0.71	1.36
['PCS3 - Inadequate training or currency']	['AD1 - Incorrect decision or plan']	0.06	0.72	1.40
['PER3 - Misperception of changing environment']	['AD1 - Incorrect decision or plan']	0.05	0.87	1.68
['OS2 - Safety risk management (proactive)', 'OS5 - Publications/procedures/written guidance']	['LT1 - Inadequate leadership or supervision']	0.05	0.68	2.29

Appendix

Table A1
Mapping of Original SHIELD Codes to SHIELD-NLP Codes

Original Code(s)	SHIELD-NLP Code
AP2 – Auditory detection, AP3 – Kinaesthetic detection, AP4 – Other senses	AP2 – No/wrong/late auditory or other detection
AR1 – Timing error, AR2 – Sequence error	AR1 – Timing or sequence error
AR3 – Right action/wrong object, AR4 – Wrong action/right object	AR3 – Action and Object Mismatch
OC1- Safety Culture, OC2 – Multi-cultural factors	OC1 – Safety culture and multi-cultural factors
OE1 – Contractors, OR1 – Personnel, OR4 – Inadequate training programme	OR1 – Personnel, contractors or inadequate training
OE3 – Economic pressure, OR2 – Budgets, OR3 – Materials availability	OR3 – Equipment/materials availability and/or budgets/economic pressures
PCO2 – Rank-based communication issues, PCO3 – Language difficulties, PCO4- Complex communication	PCO2 – Inadequate or complex communication and language difficulties
PPE2 – Movement affected, PPE3 – Hearing, PPE4 – Mental process, PPE5 – Heat/cold stress	PPE2 – Environmental stressors on human performance
PPF1 – Emotional state, PPF2 – Personality style, PPF3 – Confidence level, PPF5 – Motivation	PPF1 – Personal Factors
PME1 – Forget action/intentions, PME2 – No/inaccurate recall, PME – Negative habit	PME1- Memory
PER2 – Visual illusion, PER3 – Misperception of changing environment, PER4 – Misinterpreted or misread instrument	PER2 – Misperception and/or visual illusion
LO3 – Inappropriate pressure to perform a task, LO4 – Directed task with inadequate qualification, LO5 – Directed task with limited experience, LO6 – Directed task with inadequate equipment	LO3 – Unsafe supervisory direction/pressure to perform
LP1 – No personnel measures against regular risky behaviour, LP2 – Inappropriate behaviour affects learning, LP4 – Lack of feedback on safety reporting	LP1 - Inadequate Supervision
PEW2 – Technology create an unsafe situation, PEW5 – Communication equipment inadequate, PEW6 – Fuels or materials	PEW2 - Technology and equipment creates an unsafe situation
PPE6 - Operation more difficult due to weather and environment, PPE5 - Heat or cold stress	PPE6 - Operation more difficult due to weather and environment
PTG1 - Team working towards different goals, PTG4 - No adaptation of team performance in demanding situation	PTG1 - Team working issues
PTG2 - No cross-check and speaking-up by team member, PTG3 - No monitoring and speaking-up of team status and functioning	PTG2 - No cross-check and speaking-up by team member

Table A2
Codes removed from analysis and reasons

Code	Count	Reason Removed
AI4 – Sabotage	1	Conceptually distinct, no plausible merging category
AR5 - Lack of physical coordination		
PAW6 – unsuitable mental model	1	Unique cognitive mechanism

(continued on next page)

Table A2 (continued)

Code	Count	Reason Removed
PAW5 – Geographically lost	2	No semantic neighbour, extremely rare
PCS4 – Body/coordination limitations	1	Physiological factor; too isolated
PDN1 – Alcohol/drugs	10	Not conceptually aligned with other cognitive/perceptual codes
PMW3 – Information processing overload	1	Unique mechanism not represented elsewhere
LP3 – Personality conflict	3	Interpersonal issue; no meaningful merge
LT4 – Allowing unwritten policies to become standard	2	Rare organisational factor; too isolated
OE4 – Tempo of operations	5	Operational factor; no stable merge category
PAW7 – Pre-conceived notion or expectancy	11	Cognitive bias; insufficient support
PEW4 – Personal protective equipment inference	1	Highly specific; no merge option
PPC1 – Injury or illness existed during operation	4	Medical condition; conceptually isolated
PPC3 – Mentally exhausted (burnout)	6	Long-term psychological state; too rare
PPF4 – Performance/peer pressure	10	Social pressure; no clear merge

Table A3

Distribution of categories and codes by their number of instances in the dataset.

Category	Code	N
Communicating	AC1 - Incorrect/unclear transmission of information	72
	AC2 - No transmission of information	158
Intentional Deviation	AI1 - Workaround in normal conditions	59
	AI2 - Routine workaround	20
	AI3 - Workaround in exceptional conditions	33
Perception	AP1 - No/wrong/late visual detection	113
	AP2 - No/wrong/late auditory or other detection	25
Planning and Decision Making	AD1 - Incorrect decision or plan	251
	AD2 - Late decision or plan	48
	AD3 - No decision or plan	63
Response Execution	AR1 - Timing or sequence error	66
	AR3 - Action and Object Mismatch	35
Awareness	PAW1 - Channelized attention	24
	PAW2 - Confusion	26
	PAW3 - Distraction	38
	PAW4 - Inattention	39
Competence, skill, and capability	PCS1 - Inadequate experience	53
	PCS2 - Lack of proficiency	51
	PCS3 - Inadequate training or currency	31
Equipment and workplace	PEW1 - Ergonomics and human machine interface issues	29
	PEW2 - Technology creates an unsafe situation	87
	PEW3 - Workspace or working position incompatible with operation	32
Interpersonal communication	PCO1 - Briefing or handover inadequate	24
	PCO2 - Inadequate or complex communication and language difficulties	27
Memory/misperception	PER2 - Misperception and/or Visual illusion	34
	PME1 - Memory	31
Personal Factors	PMW1 - High workload	28
	PPC2 - Fatigue	54
	PPF1 - Personal Factors	43
	PPF7 - Risk underestimation	39
Physical Factors	PPE1 - Vision affected by environment	42
	PPE2 - Environmental stressors on human performance	35
	PPE6 - Operation more difficult due to weather and environment	45
Team/group	PTG1 - Team working issues	22
	PTG2 - No cross-check and speaking-up by team member	25
Inadequate Supervision	LP1 - Inadequate Supervision	31
	LT1 - Inadequate leadership or supervision	123
Operational Planning	LO1 - Inadequate risk assessment	108
	LO2 - Inadequate crew or team makeup or composition	43
	LO3 - Unsafe supervisory direction/pressure to perform	24
Task Leadership	LT2 - No correction of unsafe practices	44
	LT3 - No enforcement of existing rules	100
	LT5 - Directed deviation	23
Resources	OR1 - Personnel, contractors or inadequate training	42
	OR2 - Equipment/parts/materials availability and/or budgets and economic pressures	86
Economy and environment	OR5 - Design of equipment or procedures	45
	OR6 - Operational information	41
Safety management and culture	OC1 - Safety culture and multi-cultural factors	31
	OS1 - Organisation structure/policy	50
	OS2 - Safety risk management (proactive)	76
	OS3 - Safety risk assurance (reactive)	37
	OS4 - Safety promotion	30
	OS5 - Publications/procedures/written guidance	99

Table A4

Stage 1 classification performance for high-level SHIELD-NLP causal categories.

Category	N test	ROC-AUC	Precision	Recall	F1
PLANNING AND DECISION MAKING	73	0.808164	0.415385	0.369863	0.391304
SAFETY MANAGEMENT AND CULTURE	65	0.874958	0.486486	0.553846	0.517986
COMMUNICATING	46	0.849331	0.5	0.521739	0.510638
RESOURCES	40	0.646911	0.219512	0.225	0.222222
OPERATIONS PLANNING	36	0.683464	0.1875	0.166667	0.176471
TASK LEADERSHIP	34	0.669449	0.151515	0.147059	0.149254
PERSONAL FACTORS	33	0.799392	0.533333	0.484848	0.507937
INADEQUATE SUPERVISION	31	0.602321	0.235294	0.129032	0.166667
EQUIPMENT AND WORKPLACE	30	0.821362	0.5625	0.3	0.391304
PERCEPTION	28	0.856745	0.516129	0.571429	0.542373
COMPETENCE, SKILLS AND CAPABILITY	27	0.777778	0.5	0.555556	0.526316
AWARENESS	25	0.668762	0.285714	0.24	0.26087
PHYSICAL ENVIRONMENT	24	0.848901	0.8125	0.541667	0.65
INTENTIONAL DEVIATION	22	0.578961	0.076923	0.045455	0.057143
RESPONSE EXECUTION	20	0.580128	0.142857	0.1	0.117647
MEMORY/MISPERCEPTION	13	0.532063	0	0	0
INTERPERSONAL COMMUNICATION	10	0.597212	0.333333	0.1	0.153846
TEAM/GROUP	9	0.879513	0	0	0

Table A5

Stage 2 classification performance for fine-grained SHIELD-NLP causal codes.

Code	N test	ROC-AUC	Precision	Recall	F1
AD1 – Incorrect decision or plan	49	0.79	0.78	0.86	0.82
AD2 – Late decision or plan	9	0.76	0.50	0.11	0.18
AD3 – No decision or plan	15	0.85	0.50	0.33	0.40
OC1 – Safety culture and multi-cultural factors	6	0.82	0.50	0.17	0.25
OS4 – Safety promotion	10	0.80	0.00	0.00	0.00
OS2 – Safety risk management (proactive)	17	0.58	0.14	0.06	0.08
OS1 – Organisation structure/policy	11	0.50	0.00	0.00	0.00
OS5 – Publications/procedures/written guidance	15	0.77	0.50	0.47	0.48
OS3 – Safety risk assurance (reactive)	6	0.93	0.33	0.17	0.22
AC2 – No transmission of information	32	0.82	0.81	0.91	0.85
AC1 – Incorrect/unclear transmission of information	14	0.82	0.70	0.50	0.58
OR5 – Design of equipment or procedures	12	0.69	0.67	0.50	0.57
OR3 – Equipment/parts/materials availability and/or budgets and economic pressures	12	0.88	0.63	0.42	0.50
OR1 – Personnel, contractors or inadequate training	6	1.00	1.00	1.00	1.00
OR6 – Operational information	10	0.85	0.67	0.60	0.63
LO2 – Inadequate crew or team makeup or composition	10	0.86	0.64	0.90	0.75
LO1 – Inadequate risk assessment	18	0.96	0.89	0.94	0.92
LO3 – Unsafe supervisory direction/pressure to perform	6	0.69	0.00	0.00	0.00
LT3 – No enforcement of existing rules	16	0.65	0.56	0.88	0.68
LT5 – Directed deviation	6	0.84	1.00	0.33	0.50
LT2 – No correction of unsafe practices	10	0.63	0.33	0.10	0.15
PPF7 – Risk underestimation	5	0.72	0.25	0.40	0.31
PPF1 – Personal Factors	12	0.63	0.00	0.00	0.00
PPC2 – Fatigue	11	0.88	0.80	0.73	0.76
PMW1 – High workload	5	0.78	0.40	0.40	0.40
LT1 – Inadequate leadership or supervision	25	0.66	0.79	0.92	0.85
LP1 – Inadequate Supervision	6	0.73	0.00	0.00	0.00
PEW2 – Technology creates an unsafe situation	18	0.67	0.71	0.56	0.63
PEW1 – Ergonomics and human machine interface issues	7	0.63	0.40	0.29	0.33
PEW3 – Workspace or working position incompatible with operation	5	0.87	0.75	0.60	0.67
AP1 – No/wrong/late visual detection	25	1.00	1.00	1.00	1.00
AP2 – No/wrong/late auditory or other detection	3	1.00	1.00	1.00	1.00
PCS2 – Lack of proficiency	9	0.82	0.83	0.56	0.67
PCS3 – Inadequate training or currency	6	0.62	0.40	0.33	0.36
PCS1 – Inadequate experience	12	0.77	0.69	0.75	0.72
PAW4 – Inattention	7	0.63	0.00	0.00	0.00
PAW2 – Confusion	9	0.58	0.00	0.00	0.00
PAW3 – Distraction	7	0.89	0.75	0.43	0.55
PAW1 – Channelized attention	2	0.98	0.67	1.00	0.80
PPE6 – Operation more difficult due to weather and environment	10	0.96	1.00	0.50	0.67
PPE1 – Vision affected by environment	7	0.99	1.00	0.57	0.73
PPE2 – Environmental stressors on human performance	7	0.76	0.75	0.43	0.55
AI1 – Workaround in normal conditions	8	0.57	0.39	0.88	0.54
AI3 – Workaround in exceptional conditions	12	0.63	0.33	0.08	0.13
AI2 – Routine workaround	2	0.80	0.00	0.00	0.00
AR1 – Timing or sequence error	12	0.51	0.60	0.75	0.67
AR3 – Action and Object Mismatch	8	0.48	0.29	0.25	0.27
PER2 – Misperception and/or Visual illusion	5	0.89	0.57	0.80	0.67

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Table A5 (continued)

Code	N test	ROC-AUC	Precision	Recall	F1
PME1 – Memory	8	0.89	0.83	0.63	0.71
PCO1 – Briefing or handover inadequate	4	0.85	0.75	0.75	0.75
PCO2 – Inadequate or complex communication and language difficulties	6	0.85	0.83	0.83	0.83
PTG2 – No cross-check and speaking-up by team member	4	0.40	0.43	0.75	0.55
PTG1 – Team working issues	5	0.40	0.50	0.20	0.29

Data availability

The data utilised in this study are confidential. Access to the dataset can only be obtained through direct contact with the relevant stakeholders of the SAFEMODE project.

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